

Cointegration and Pair Trading

Empirical Method for Finance

Group 7

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Abstract

This paper explores the theory and practice of cointegration-based pair trading in equity markets. We begin by examining the statistical foundations of cointegration, focusing on how non-stationary asset price series can form a stationary linear combination—an essential requirement for pairs trading. Through unit-root tests (Dickey–Fuller) on daily log prices for several major energy sector stocks, we find that most exhibit non-stationarity ($I(1)$), except for **TOTALENERGIES**, which appears stationary. This outcome affects subsequent cointegration analyses, as stationarity is typically assumed for individual series entering a cointegration relationship. Next, we employ Monte Carlo simulations to derive sample-specific critical values for the Dickey–Fuller statistic on both individual time series and residuals. We then identify a statistically significant cointegration relationship between **CHEVRON** and **CONOCOPHILLIPS**, evident in both log-price and raw-price regressions. We confirm that the resulting spread is indeed mean-reverting, although it displays a high degree of persistence and slow mean reversion. Building on this evidence, we design and implement a pair trading strategy. We outline its signal generation (based on the normalized spread crossing predefined thresholds), leverage management, and stop-loss rules. The empirical results illustrate how leveraging the pair’s mean-reverting spread can be profitable—under controlled leverage and adequate risk management. However, they also highlight the practical challenges of prolonged drawdowns and elevated carrying costs when the spread reverts slowly. Overall, our findings underscore both the theoretical appeal of cointegration-based pair trading and the operational complexities of deploying it in real markets. By integrating robust statistical tests, realistic simulation-based critical values, and a comprehensive risk framework, we provide a practical roadmap for traders seeking to exploit persistent yet slowly mean-reverting deviations between fundamentally related assets.

Keywords: Cointegration, Pair Trading, Stationarity, Unit Root Testing (Dickey–Fuller), Energy Sector Stocks, Mean Reversion, Monte Carlo Simulations, Non-stationary Time Series, Spread Trading, Risk Management, Leverage Constraints

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1 Descriptive Statistics

Q1.1

		EXXON_MOBIL	CONOCOPHILLIPS	CHEVRON	TOTALENERGIES	OCCIDENTAL_PTL
Daily Simple Returns	annualized_mean	0.0602	0.1141	0.0787	0.0666	0.0574
	annualized_variance	0.0605	0.1094	0.0694	0.1308	0.1717
	skewness	0.0594	0.0609	-0.2287	1.0226	-1.0456
	kurtosis	10.3903	18.8826	27.5277	32.7867	61.8794
	min	-0.1222	-0.2484	-0.2212	-0.2163	-0.5343
	max	0.1269	0.2521	0.2274	0.3187	0.3370
Daily Log Returns	annualized_mean	0.0299	0.0593	0.0438	0.0017	-0.0327
	annualized_variance	0.0606	0.1100	0.0701	0.1295	0.1872
	skewness	-0.1594	-0.5225	-0.9150	0.0654	-5.0656
	kurtosis	10.4856	20.8544	30.4959	27.8820	167.8387
	min	-0.1304	-0.2856	-0.2501	-0.2437	-0.7641
	max	0.1194	0.2249	0.2049	0.2766	0.2904

Table 1: Comparison between daily simple and log returns (252 trading days)

We can see in Table 1 that daily simple returns tend to overstate performance when compounded over time, whereas daily log-returns are additive across periods and better suited to many statistical models (e.g., they often approximate normality more closely). In the table, we observe higher annualized means for simple returns but slightly higher extremes (skewness, kurtosis) for log-returns, reflecting heavier tails and stronger deviations from normality. Moreover, minimum and maximum values are typically more pronounced under log-returns, highlighting their stronger sensitivity to large price moves.

Q1.2

		EXXON_MOBIL	CONOCOPHILLIPS	CHEVRON	TOTALENERGIES	OCCIDENTAL_PTL
Weekly Simple Returns	annualized_mean	0.0585	0.1155	0.0756	0.0381	0.0529
	annualized_variance	0.0538	0.1048	0.0606	0.0783	0.1660
	skewness	-0.1166	-0.1224	0.1442	-0.0999	0.6975
	kurtosis	5.8290	7.5764	6.2260	6.7570	31.1996
	min	-0.1538	-0.2751	-0.1682	-0.2207	-0.5653
	max	0.1313	0.2122	0.1579	0.1839	0.5715
Weekly Log Returns	annualized_mean	0.0315	0.0626	0.0454	-0.0013	-0.0326
	annualized_variance	0.0542	0.1065	0.0605	0.0793	0.1783
	skewness	-0.3556	-0.6144	-0.1263	-0.4509	-2.7488
	kurtosis	6.1748	9.2787	6.3961	7.4556	60.1049
	min	-0.1670	-0.3218	-0.1841	-0.2494	-0.8332
	max	0.1233	0.1924	0.1466	0.1688	0.4520

Table 2: Comparison between weekly simple and log returns

We can see in Table 2 that on a weekly basis, we observe the same phenomenon as with daily returns: simple returns generally show higher annualized means, while log-returns exhibit more pronounced extremes (kurtosis) and sometimes more negative skewness (notably for Occidental). Moreover, minimum values tend to be lower and maximum values slightly smaller for log-returns, reflecting the compression effect of the logarithm. Overall, simple returns can therefore overestimate long-term performance, whereas log-returns remain more coherent with volatility models and the additive property of returns over multiple periods.

Q1.3

When moving from daily to weekly log-returns, the distribution of returns tends to become slightly “smoother,” often showing lower (or differently shifted) skewness and reduced kurtosis, because large daily price movements get aggregated over a longer interval. Annualized means and variances can also shift, with weekly log-returns frequently displaying a smaller annualized variance if volatility is less amplified over a single weekly interval. In other words, converting daily data to a weekly frequency generally reduces the impact of extreme daily fluctuations and can alter key descriptive statistics—particularly skewness, kurtosis, and variance.

2 Stationarity

The first step in implementing a pair trading strategy is to test whether each time series of log-prices is stationary. Stationarity is important because it guarantees that statistical properties (mean, variance, and autocorrelations) do not change over time, enabling us to apply standard inference and forecasting methods.

To do so, we fit an AR(1) model (i.e., a one-lag autoregression) to each log-price series:

$$p_t = \mu + \phi p_{t-1} + \varepsilon_t,$$

where ε_t is a mean-zero error term. We then use the Dickey-Fuller¹ (DF) test on the estimated model to assess whether $\{p_t\}$ is stationary (i.e., $|\phi| < 1$) or exhibits a unit root ($\phi = 1$).

Q2.1 Dickey-Fuller Test: Hypotheses & Intuition

We consider the following regression:

$$p_t = \mu + \phi p_{t-1} + \varepsilon_t, \quad \text{where } p_t = \log(P_t).$$

- **Null hypothesis** (H_0): $\phi = 1$.
 - Under H_0 , p_t follows a *random walk* (unit root) $\Rightarrow$ non-stationary.
- **Alternative hypothesis** (H_1): $\phi < 1$.
 - Under H_1 , p_t is a stationary AR(1) process (transitory shocks, constant variance).

Intuition: If $\phi = 1$, any shock ε_t has a *permanent* effect on p_t , causing the log-price series to drift over time with a rising variance — which violates stationarity. Conversely, if $\phi < 1$, the process exhibits mean reversion and a bounded variance (exponentially decaying autocorrelations).

Q2.2 DF Test Statistic and Decision Rule

To test $H_0 : \phi = 1$, we estimate the regression (via OLS) and compute the DF statistic:

$$t(\hat{\phi} - 1) = \frac{\hat{\phi} - 1}{\widehat{\text{Std}}(\hat{\phi})},$$

where

$$\widehat{\text{Var}}(\hat{\phi}) \text{ is derived from } \hat{\phi}_{\varepsilon}^2 = \frac{1}{T-1} \sum_{t=1}^T (p_t - \hat{\mu} - \hat{\phi} \cdot p_{t-1})^2.$$

The critical values for this test do *not* follow the standard Student's t distribution and must come from DF tables or be obtained via simulation. This non-standard behavior arises because under the null hypothesis the regressor p_{t-1} is non-stationary, so the usual Central Limit Theorem does not apply. . It is a **left-tailed** test:

- **Rejection rule:**
 - If $t(\hat{\phi} - 1)$ is more negative (i.e., smaller) than the DF critical value (e.g., -2.89 at 5%), we **reject** H_0 and conclude $\phi < 1$ (stationarity).
 - Otherwise, we fail to reject the unit-root hypothesis and treat p_t as non-stationary.

Example: Suppose we compute $t(\hat{\phi} - 1) = -3.1$ and the 5% critical value is -2.89 . Since $-3.1 < -2.89$, we reject H_0 and conclude that the series is stationary.

Q2.3 Equivalence of Difference Regression

We can also write the model in differences:

$$r_t = p_t - p_{t-1} = \mu + \phi p_{t-1} + \varepsilon_t.$$

Here, $\phi = 0$ in the *difference form* corresponds to $\phi = 1$ in the *level form*. Indeed:

$$\underbrace{p_t - p_{t-1}}_{r_t} = \mu + \phi p_{t-1} + \varepsilon_t \iff p_t = (\phi + 1) p_{t-1} + \mu + \varepsilon_t.$$

- In differences: $H_0: \phi = 0 \Rightarrow p_t$ non-stationary.
- In levels: $H_0: \phi = 1 \Rightarrow p_t$ non-stationary.

Hence, testing $\phi = 0$ in the difference regression is equivalent to testing $\phi = 1$ in the original AR(1) model. Both detect the presence or absence of a **unit root**.

2.1 Critical Values

Q2.1.1

Simulation of error terms : We generate $N = 10\,000$ replications of a time series of $T = 3941$ (where 3941 is our sample length) error terms $\varepsilon_t^{(i)}$, where each term is drawn from the standard normal distribution $N(0, 1)$. This yields an array of simulated innovations that will drive the random walk model.

Q2.1.2

Simulation of Log-Price Series : Using the simulated error terms, we compute each log-price series under the null hypothesis by assuming a random walk:

$$p_t^{(i)} = p_{t-1}^{(i)} + \varepsilon_t^{(i)},$$

with an initial value $p_0^{(i)}$ set to zero. This produces N simulated trajectories of log-prices.

Q2.1.3

Estimation of the AR(1) Model and Computation of the Test Statistic: For each simulated log-price series, we estimate the AR(1) model

$$p_t^{(i)} = \mu^{(i)} + \phi^{(i)} p_{t-1}^{(i)} + \varepsilon_t^{(i)}$$

using ordinary least squares (OLS). We then calculate the test statistic

$$t(\hat{\phi}^{(i)} - 1) = \frac{\hat{\phi}^{(i)} - 1}{\widehat{\text{Std}}(\hat{\phi}^{(i)})},$$

for each replication. The distribution of these t -statistics is then used to extract the critical values at the 10%, 5%, and 1% significance levels.

This simulation process provides us with sample-size-specific critical values for the DF test, which is necessary because the test statistic does not follow the standard Student's t distribution due to the non-stationarity of the lagged log-price variable.

Q2.1.4

Under the null hypothesis of the DF test, the log-price series $p_t = \log(P_t)$ is assumed to follow a random walk, i.e.,

$$p_t = p_{t-1} + \varepsilon_t,$$

where the error terms ε_t are independent and identically distributed as $N(0, 1)$ (white noise). This assumption implies that the series contains a unit root and is non-stationary. Consequently, the usual conditions for the Central Limit Theorem do not hold, and the test statistic

$$t(\hat{\phi} - 1)$$

follows a non-standard distribution rather than the standard Student's t distribution.

By simulating random walks using Monte Carlo² experiments, we generate an empirical distribution of $t(\hat{\phi} - 1)$ under the null hypothesis. This approach allows us to:

1. Capture the true sampling variability of the test statistic for our specific sample size T .
2. Obtain accurate critical values (e.g., the 10%, 5%, and 1% quantiles) tailored to our empirical setting.

In summary, simulating random walks is essential because it provides a realistic, sample-size-specific benchmark for evaluating the DF test statistic, ensuring our inference on stationarity is reliable.

Q2.1.5

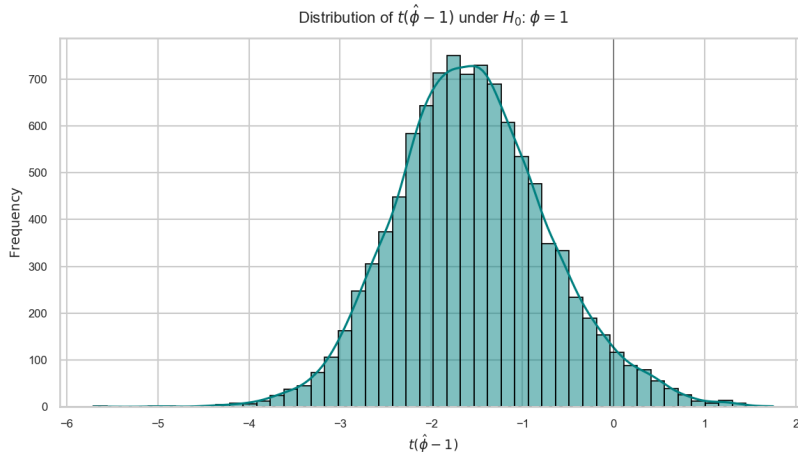


Figure 1: Histogram of the simulated $t(\hat{\phi} - 1)$ statistics (for $N = 10'000$, $T = 3941$, seed=123).

Figure 1 illustrates the histogram of the N values of $t(\hat{\phi} - 1) = \frac{\hat{\phi} - 1}{\widehat{\text{Std}}(\hat{\phi})}$, obtained from our Monte Carlo² simulations under the null hypothesis $H_0 : \phi = 1$.

Observations:

- The distribution is markedly non-standard and deviates from the conventional Student's t -distribution.
- It exhibits skewness and heavier tails, reflecting the non-stationarity of the regressor p_{t-1} in the AR(1) model.
- The empirical quantiles of this distribution (e.g., the 10%, 5%, and 1% percentiles) are used as critical values for the DF test.

Interpretation: The distribution of $t(\hat{\phi} - 1)$ represents the sampling distribution of the DF test statistic under the null hypothesis of a unit root. Because the regressor p_{t-1} is non-stationary, the usual Central Limit Theorem does not apply, leading to a non-standard asymptotic distribution. Therefore, critical values must be determined via simulation rather than using standard t -distribution tables.

Q2.1.6

	10%	5%	1%
Critical Values	-2.5682	-2.8759	-3.4461

Table 3: Critical values (10%, 5%, 1%) of the DF test from the simulated t-statistics distribution (seed=123).

Q2.1.7

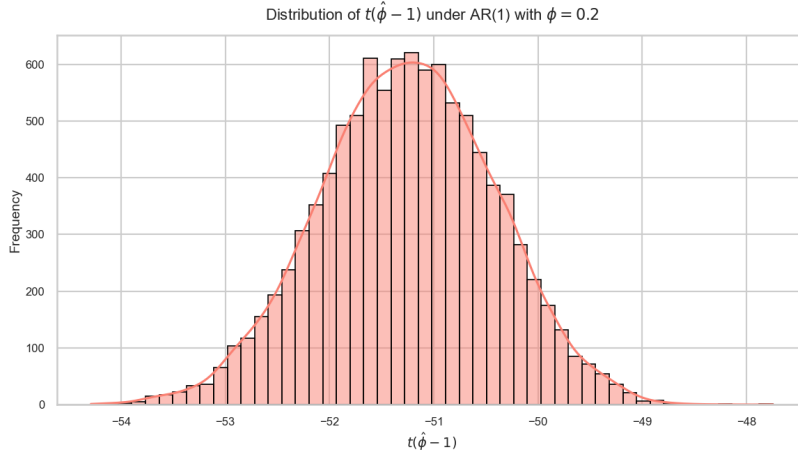


Figure 2: Histogram of the simulated $t(\hat{\phi} - 1)$ statistics with $\phi = 0.2$ (for $N = 10'000$, $T = 3941$, seed=123).

When we redo the simulation under the stationary AR(1) model : $p_t^{(i)} = 0.2 \cdot p_{t-1}^{(i)} + \varepsilon_t^{(i)}$,

the distribution of $t(\hat{\phi}^{(i)} - 1)$ shifts dramatically to the left, as seen in Figure 2. This occurs because the true value $\phi = 0.2$ is well below 1, so the estimator $\hat{\phi}^{(i)}$ converges near 0.2 rather than 1. Consequently, the numerator $\hat{\phi}^{(i)} - 1$ is markedly negative, while the standard error of $\hat{\phi}^{(i)}$ remains relatively small for large sample sizes. The result is a distribution centered on very negative t -statistics, confirming that the process is far from having a unit root and is indeed stationary.

Simulation for $T = 500$

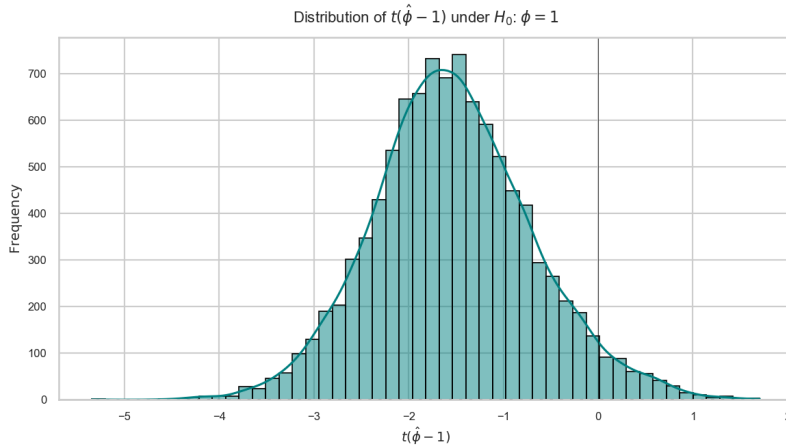


Figure 3: Histogram of the simulated $t(\hat{\phi} - 1)$ statistics (for $N = 10'000$, $T = 500$, seed=123).

	10%	5%	1%
Critical Values for T=500	-2.5398	-2.8477	-3.3647

Figure 4: Critical values (10%, 5%, 1%) for DF test (seed=123)

The empirical distribution of these t -statistics was then used to determine the 10%, 5%, and 1% critical values. Figure 3 displays the histogram of $t(\hat{\phi} - 1)$ for $T = 500$, and Table 4 reports the corresponding critical values.

2.2 Testing Non-Stationarity

Q2.8

Our analysis is based on the daily closing prices of the five assets: EXXON_MOBIL, CONOCOPHILLIPS, CHEVRON, TOTALENERGIES, and OCCIDENTAL_PTL. For each asset, we compute the log-price series (i.e., $p_t = \log(P_t)$) and estimate the following AR(1) regression:

$$p_t = \mu + \phi p_{t-1} + \varepsilon_t.$$

The DF test statistic is then computed as:

$$t(\hat{\phi} - 1) = \frac{\hat{\phi} - 1}{\widehat{\text{Std}}(\hat{\phi})}.$$

Using the simulated distribution of $t(\hat{\phi} - 1)$ from our earlier analysis (Q2.1.5), we obtain the corresponding critical values (10%, 5%, and 1% levels) and compute the p-value as the proportion of simulated t-statistics that are lower than the observed value (given the left-tailed nature of the test).

	DF_statistic	Critical_value (10%)	p_value	Conclusion
EXXON_MOBIL	-1.8347	-2.5682	0.3629	Fail to reject H_0 : Unit root present
CONOCOPHILLIPS	-1.9408	-2.5682	0.3139	Fail to reject H_0 : Unit root present
CHEVRON	-2.5651	-2.5682	0.1007	Fail to reject H_0 : Unit root present
TOTALENERGIES	-4.6598	-2.5682	0.0001	Reject H_0 : Series is stationary
OCCIDENTAL_PTL	-1.7945	-2.5682	0.3846	Fail to reject H_0 : Unit root present

Table 4: DF test results for the five assets (critical value at 10% level).

	DF_statistic	Critical_value (5%)	p_value	Conclusion
EXXON_MOBIL	-1.8347	-2.8759	0.3629	Fail to reject H_0 : Unit root present
CONOCOPHILLIPS	-1.9408	-2.8759	0.3139	Fail to reject H_0 : Unit root present
CHEVRON	-2.5651	-2.8759	0.1007	Fail to reject H_0 : Unit root present
TOTALENERGIES	-4.6598	-2.8759	0.0001	Reject H_0 : Series is stationary
OCCIDENTAL_PTL	-1.7945	-2.8759	0.3846	Fail to reject H_0 : Unit root present

Table 5: DF test results for the five assets (critical value at 5% level).

	DF_statistic	Critical_value (1%)	p_value	Conclusion
EXXON_MOBIL	-1.8347	-3.4461	0.3629	Fail to reject H_0 : Unit root present
CONOCOPHILLIPS	-1.9408	-3.4461	0.3139	Fail to reject H_0 : Unit root present
CHEVRON	-2.5651	-3.4461	0.1007	Fail to reject H_0 : Unit root present
TOTALENERGIES	-4.6598	-3.4461	0.0001	Reject H_0 : Series is stationary
OCCIDENTAL_PTL	-1.7945	-3.4461	0.3846	Fail to reject H_0 : Unit root present

Table 6: DF test results for the five assets (critical value at 1% level).

The test results indicate that:

- For EXXON_MOBIL, CONOCOPHILLIPS, CHEVRON, and OCCIDENTAL_PTL, the computed DF statistics (approximately -1.83 , -1.94 , -2.57 , and -1.79 , respectively) are not sufficiently negative compared to the critical values (e.g., -2.5852 at the 10% level, -2.8613 at the 5% level, and -3.4421 at the 1% level). Their high p-values (ranging from about 0.0986 to 0.3793) lead us to **fail to reject** the null hypothesis of a unit root. Thus, these log-price series are deemed non-stationary.
- In contrast, TOTALENERGIES has a DF statistic of approximately -4.66 , which is well below all the critical thresholds. With a p-value of 0.0002, we **reject** the null hypothesis at all conventional significance levels, indicating that the log-price series for TOTALENERGIES is stationary.

Q2.9

The results from our DF tests have important implications for cointegration analysis. In cointegration studies, we generally assume that each asset's log-price series is integrated of order one (I(1)), meaning that they are non-stationary in levels, and that a suitable linear combination of these I(1) series can be stationary (I(0)).

Our findings show that four out of the five assets (EXXON_MOBIL, CONOCOPHILLIPS, CHEVRON, and OCCIDENTAL_PTL) exhibit a unit root and are thus I(1). This behavior is what we typically observe in financial markets, supporting the use of cointegration techniques for these series.

However, the log-price series for TOTALENERGIES appears to be stationary (I(0)), which is unusual. Mixing a stationary series with non-stationary series when testing for cointegration can lead to misleading or spurious results. Therefore, although most of the assets are suitable for cointegration analysis, the case of TOTALENERGIES should be carefully examined before including it in a cointegrated portfolio.

In summary, while the majority of the log-price series are non-stationary—aligning with common empirical observations—the exception of TOTALENERGIES suggests a different underlying dynamic that requires further investigation.

3 Cointegration

In pair trading, it is essential to determine whether two asset price series are cointegrated. Cointegration implies that although the individual log-price series may be non-stationary, a linear combination of them is stationary, suggesting the existence of a long-run equilibrium relationship. To test for cointegration between two assets, we first estimate the relationship

$$p_t^A = \alpha + \beta p_t^B + z_t,$$

where z_t represents the residuals. Under the null hypothesis of no cointegration, z_t is non-stationary. We then test for a unit root in z_t by estimating

$$\Delta \hat{z}_t = \mu + \phi \hat{z}_{t-1} + \varepsilon_t.$$

The t-statistic for testing $H_0 : \phi = 0$ is computed, and its distribution under the null is obtained via Monte Carlo simulation. In Subsection 3.1, we detail the simulation procedure for a sample size of $T = 500$ to derive the corresponding critical values.

Q3.1 Critical Values for the Cointegration Test (Simulation)

To evaluate the cointegration relationship between any two assets, we first simulate the distribution of the test statistic under the null hypothesis of no cointegration. This is done as follows:

1. Simulation of independent random walks: For a given sample size $T = 500$, we simulate two independent random walks for assets A and B. Specifically, we generate two series of error terms $\varepsilon_t^{A(i)}$ and $\varepsilon_t^{B(i)}$, each drawn from $N(0, 1)$, and compute the corresponding log-price series using

$$p_t^{A(i)} = p_{t-1}^{A(i)} + \varepsilon_t^{A(i)} \quad \text{and} \quad p_t^{B(i)} = p_{t-1}^{B(i)} + \varepsilon_t^{B(i)},$$

with the initial value $p_0 = 0$. Figure 5 shows the histogram of the simulated t-statistics $t(\hat{\phi})$ for the residuals. The distribution is non-standard, as expected under the null hypothesis of no cointegration (i.e., when both asset price series are independent random walks).

2. Estimation of the cointegration regression: For each replication i , we estimate the cointegration relationship

$$p_t^{A(i)} = \alpha + \beta p_t^{B(i)} + z_t^{(i)},$$

and compute the residual series $\hat{z}_t^{(i)}$.

From the simulated distribution, we compute the empirical critical values corresponding to the 10%, 5%, and 1% significance levels shown in Table 7. These values represent the thresholds below which the test statistic must fall to reject the null hypothesis of no cointegration. In other words, if the t-statistic computed from the actual residuals of a cointegration regression is more negative than the appropriate critical value, we reject the null hypothesis and conclude that the asset pair is cointegrated.

3. Dickey–Fuller Test on the residuals: We then test the null hypothesis that the residual series $\hat{z}_t^{(i)}$ is non-stationary by estimating the regression

$$\Delta \hat{z}_t^{(i)} = \mu + \phi \hat{z}_{t-1}^{(i)} + \varepsilon_t^{(i)},$$

and computing the test statistic $t(\hat{\phi}^{(i)})$ (i.e., the t-statistic for testing $H_0 : \phi = 0$).

Q3.1 Analysis of the Simulated Distribution and Critical Values

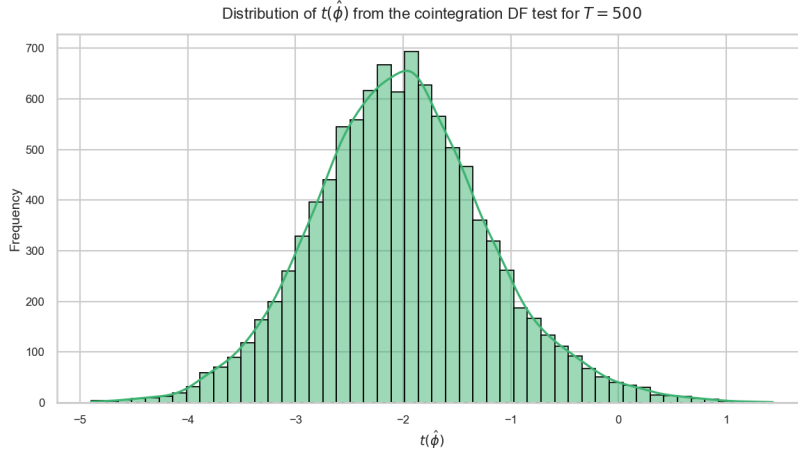


Figure 5: Histogram of the simulated $t(\hat{\phi})$ statistics from the cointegration DF test (for $N = 10'000$, $T = 500$, seed=123).

Figure 5 displays the histogram of the simulated $t(\hat{\phi})$ statistics obtained from the cointegration Dickey–Fuller test, based on $N = 10\,000$ replications with $T = 500$. From this histogram, we observe that the distribution of the test statistic is clearly non-standard—it is notably left-skewed with a heavy left tail. This behavior reflects the fact that under the null hypothesis of no cointegration (i.e., when both asset price series follow independent random walks), the residuals exhibit a unit root and the standard asymptotic results do not hold.

The empirical critical values calculated from the simulated distribution are:

	10%	5%	1%
Critical Values for T=500	-3.0211	-3.3290	-3.8781

Table 7: critical values (10%, 5%, 1%) of the DF test from the simulated t-statistics distribution (seed=123).

These values represent the thresholds below which the test statistic must fall to reject the null hypothesis of no cointegration. In practice, if the t-statistic computed from the cointegration regression on the actual asset pairs is more negative than the corresponding critical value (e.g., below -3.3290 at the 5% level), we would reject the null and conclude that the residual series is stationary, implying that the asset pair is cointegrated.

Overall, the non-standard nature of the simulated distribution—evidenced by its left skew and heavy tail—justifies the use of Monte Carlo simulation for obtaining critical values tailored to our sample size ($T = 500$) rather than relying on conventional t-distribution tables.

3.2 Testing for Cointegration

Before conducting cointegration tests, it is essential to review the stationarity properties of the individual asset log-price series. As shown in the previous DF test results, the log-price series for EXXON_MOBIL, CONOCOPHILLIPS, CHEVRON, and OCCIDENTAL_PTL fail to reject the null hypothesis of a unit root (i.e., they are non-stationary or $I(1)$). However, the series for TOTALENERGIES is found to be stationary ($I(0)$) since its test statistic is significantly lower than the critical values and the null is rejected (cf. Q2.8 : Table 4, Table 5, Table 6).

Since the cointegration framework requires that each individual series be integrated of order 1 ($I(1)$) so that a linear combination can potentially be stationary ($I(0)$), the fact that TOTALENERGIES is already stationary poses a methodological challenge. In practice, it is not appropriate to test for cointegration between a stationary series and a non-stationary series because the underlying assumptions of the cointegration analysis would be violated. Consequently, when selecting pairs for cointegration analysis (and thus for our pair trading strategy), we decided to exclude TOTALENERGIES, ensuring that only pairs of assets with similar integration properties ($I(1)$) are compared.

In the following subsections, we report the cointegration test results for each pair (in both directions) among the remaining assets, and we comment on the estimated parameters as well as the economic intuition behind the strongest cointegration relationship.

Q3.2

Asset_A	Asset_B	Direction	$(\hat{\alpha})$	$(\hat{\beta})$	DF_Statistic	Critical_5%	p_value	Conclusion
CHEVRON	CONOCOPHILLIPS	CHEVRON $\rightarrow$ CONOCOPHILLIPS	2.3391	0.5783	-3.7674	-3.3290	0.0149	Reject H_0 : Cointegration exists
CONOCOPHILLIPS	CHEVRON	CONOCOPHILLIPS $\rightarrow$ CHEVRON	-2.7543	1.4566	-3.3732	-3.3290	0.0444	Reject H_0 : Cointegration exists
CHEVRON	EXXON_MOBIL	CHEVRON $\rightarrow$ EXXON_MOBIL	1.9173	0.6414	-2.5536	-3.3290	0.2499	Fail to reject H_0 : No cointegration
CHEVRON	OCCIDENTAL_PTL	CHEVRON $\rightarrow$ OCCIDENTAL_PTL	4.3547	0.0919	-2.4065	-3.3290	0.3148	Fail to reject H_0 : No cointegration
CONOCOPHILLIPS	EXXON_MOBIL	CONOCOPHILLIPS $\rightarrow$ EXXON_MOBIL	-0.2275	0.9948	-1.9069	-3.3290	0.5673	Fail to reject H_0 : No cointegration
EXXON_MOBIL	CHEVRON	EXXON_MOBIL $\rightarrow$ CHEVRON	0.4005	0.8428	-1.7864	-3.3290	0.6310	Fail to reject H_0 : No cointegration
EXXON_MOBIL	CONOCOPHILLIPS	EXXON_MOBIL $\rightarrow$ CONOCOPHILLIPS	2.2411	0.5190	-1.7842	-3.3290	0.6322	Fail to reject H_0 : No cointegration
CONOCOPHILLIPS	OCCIDENTAL_PTL	CONOCOPHILLIPS $\rightarrow$ OCCIDENTAL_PTL	3.4806	0.1602	-1.7202	-3.3290	0.6609	Fail to reject H_0 : No cointegration
EXXON_MOBIL	OCCIDENTAL_PTL	EXXON_MOBIL $\rightarrow$ OCCIDENTAL_PTL	2.8467	0.3755	-1.6292	-3.3290	0.7010	Fail to reject H_0 : No cointegration
OCCIDENTAL_PTL	EXXON_MOBIL	OCCIDENTAL_PTL $\rightarrow$ EXXON_MOBIL	-1.9783	1.3865	-1.5676	-3.3290	0.7256	Fail to reject H_0 : No cointegration
OCCIDENTAL_PTL	CHEVRON	OCCIDENTAL_PTL $\rightarrow$ CHEVRON	1.9965	0.4459	-1.5573	-3.3290	0.7300	Fail to reject H_0 : No cointegration
OCCIDENTAL_PTL	CONOCOPHILLIPS	OCCIDENTAL_PTL $\rightarrow$ CONOCOPHILLIPS	2.8297	0.3086	-1.5534	-3.3290	0.7310	Fail to reject H_0 : No cointegration

Table 8: Cointegration test results for selected asset pairs. For each ordered pair, the table displays the estimated cointegration regression parameters ($\hat{\alpha}$ and $\hat{\beta}$), the Dickey–Fuller test statistic computed on the residuals, the simulated 5% critical value, the corresponding p-value, and the conclusion regarding cointegration. Results are based on a simulated t-statistics distribution (seed = 123).

Q3.3

The estimated parameters $\hat{\alpha}$ and $\hat{\beta}$ for the cointegration regressions are reported in Table 8. For example, in the regression of CHEVRON on CONOCOPHILLIPS, the estimated intercept is $\hat{\alpha} = 2.3391$ and the slope coefficient is $\hat{\beta} = 0.5783$. This indicates that the log-price of CHEVRON increases by approximately 0.5783 units for each one-unit increase in the log-price of CONOCOPHILLIPS, after accounting for the constant term. Similar interpretations apply to the other pairs listed in the table. These parameter estimates provide crucial insights into the long-run equilibrium relationships among the assets, which are fundamental for constructing a pair trading strategy.

The cointegration relationship is expressed as

$$p_t^{\text{CHEVRON}} = 2.3391 + 0.5783 p_t^{\text{CONOCOPHILLIPS}} + z_t,$$

so that the spread is defined as

$$z_t = p_t^{\text{CHEVRON}} - 2.3391 - 0.5783 p_t^{\text{CONOCOPHILLIPS}}.$$

Note that the spread is computed using log-prices to ensure that the units are consistent across the analysis.

Q3.4

Among all tested pairs, the cointegration test results indicate that the pair CHEVRON $\rightarrow$ CONOCOPHILLIPS is the most strongly cointegrated, as it exhibits the most negative DF statistic (-3.7674) and the lowest p-value (0.0149), suggesting a robust long-run equilibrium relationship; this likely reflects that both companies are subject to similar market forces and industry dynamics, causing their prices to move together over time.

Q3.5

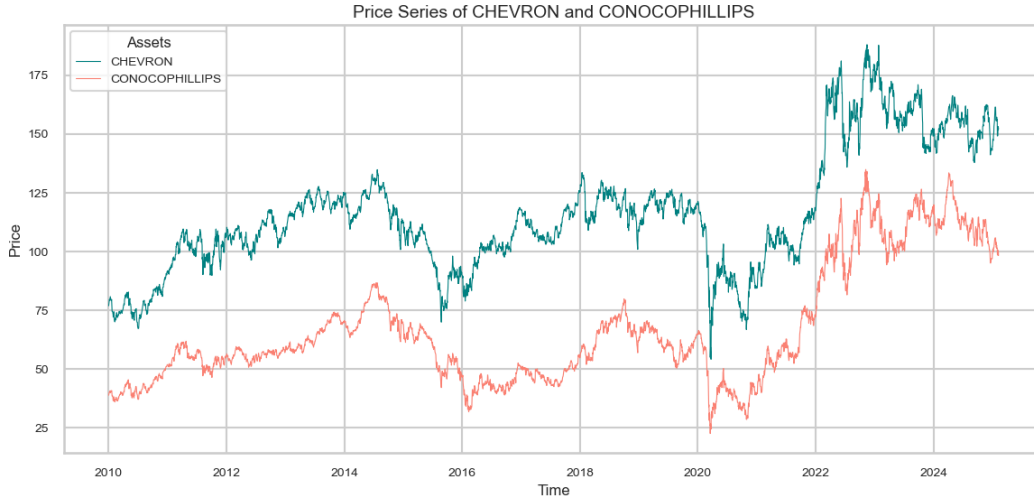


Figure 6: Raw Price series of CHEVRON and CONOCOPHILLIPS. Both assets exhibit broadly similar movements over time, reflecting common industry factors and supporting the strong cointegration result found in our tests.

In Figure 6, we see that although there are temporary divergences, the two raw price series do not drift apart indefinitely, which is consistent with the notion that they share a long-run equilibrium relationship. This visual evidence further supports the statistical finding that CHEVRON and CONOCOPHILLIPS are cointegrated, making them a suitable candidate for a pair trading strategy.

4 Pair Trading

4.1

In Section 4, we implement our pair trading strategy using raw prices rather than log-prices. This approach is chosen to simplify the construction and interpretation of the trading signal. Accordingly, we re-estimate the cointegration regression for the selected pair using the raw price series. The cointegration relationship is modeled as

$$P_t^A = \alpha + \beta P_t^B + z_t,$$

where P_t^A and P_t^B represent the raw prices of assets A and B respectively, and z_t denotes the spread. The spread is defined as

$$z_t = P_t^A - \alpha - \beta P_t^B.$$

Table 9 reports the estimated cointegration regression parameters ($\hat{\alpha}$ and $\hat{\beta}$) based on the raw prices, the Dickey–Fuller test statistic computed on the residuals, the simulated 5% critical value (Table 7), the corresponding p-value, and the conclusion regarding cointegration for each ordered pair. For example, in the regression of CHEVRON on CONOCOPHILLIPS, the estimated parameters are $\hat{\alpha} = 52.1060$ and $\hat{\beta} = 0.9620$. This implies that, after adjusting for the constant term, an increase of one unit in the raw price of CONOCOPHILLIPS is associated with an increase of approximately 0.9620 units in the raw price of CHEVRON. These results, together with the statistical significance of the test, provide a robust basis for our pair trading strategy. We note that the p-value (0.0106) obtained from the raw price analysis is more significant than that obtained from the log-price analysis (0.0149), which reinforces the robustness of the cointegration relationship for the pair under study.

Asset_A	Asset_B	Direction	Alpha	Beta	DF_Statistic	Critical_5%	p_value	Conclusion
CHEVRON	CONOCOPHILLIPS	CHEVRON → CONOCOPHILLIPS	52.1060	0.9620	-3.8568	-3.3290	0.0106	Reject H_0 (cointegration)
CONOCOPHILLIPS	CHEVRON	CONOCOPHILLIPS → CHEVRON	-37.3830	0.8951	-3.5183	-3.3290	0.0301	Reject H_0 (cointegration)

Table 9: Cointegration test results for the winner asset pairs shown in Table 8. For each ordered pair, the table displays the estimated cointegration regression parameters ($\hat{\alpha}$ and $\hat{\beta}$) that are conform the Raw Prices, the Dickey–Fuller test statistic computed on the residuals, the simulated 5% critical value, the corresponding p-value, and the conclusion regarding cointegration. Results are based on a simulated t-statistics distribution (Table 8, seed = 123).

Q4.1

Given that the pair of assets (A, B) are cointegrated, the spread

$$z_t = P_t^A - \alpha - \beta P_t^B$$

is stationary. This means that even though the individual price series P_t^A and P_t^B are non-stationary and may exhibit trends, their linear combination oscillates around a constant mean with a stable variance over time. In practical terms, if $z_t \gg 0$, it indicates that asset A is significantly overpriced relative to asset B ; hence, one would expect that asset A 's price will eventually revert downward, or asset B 's price will rise, to restore equilibrium. Conversely, if $z_t \ll 0$, it implies that asset A is underpriced relative to asset B , suggesting that asset A is likely to appreciate. This predictable mean-reverting behavior of z_t provides the basis for a statistical arbitrage strategy—specifically, pair trading—where one shorts the overvalued asset and goes long on the undervalued asset to profit from the eventual reversion.

Q4.2

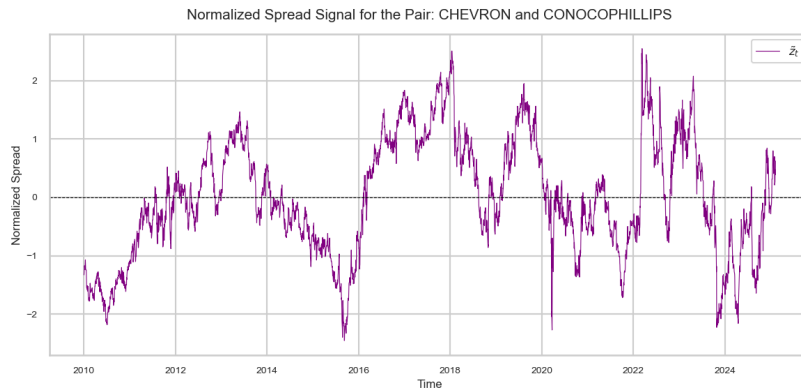


Figure 7: Normalized spread for the pair CHEVRON and CONOCOPHILLIPS.

Figure 7 displays the normalized spread signal defined as $\tilde{z}_t = \frac{z_t}{\sigma(z_t)}$, for the cointegrated pair CHEVRON and CONOCOPHILLIPS. The plot demonstrates that the spread oscillates around a fixed mean = 0, confirming its stationarity. Notably, when $\tilde{z}_t$ is significantly positive, it suggests that CHEVRON is relatively overpriced compared to CONOCOPHILLIPS, reinforcing the trading strategy of shorting CHEVRON and going long on CONOCOPHILLIPS with the expectation of a reversion towards equilibrium.

Q4.3

1.

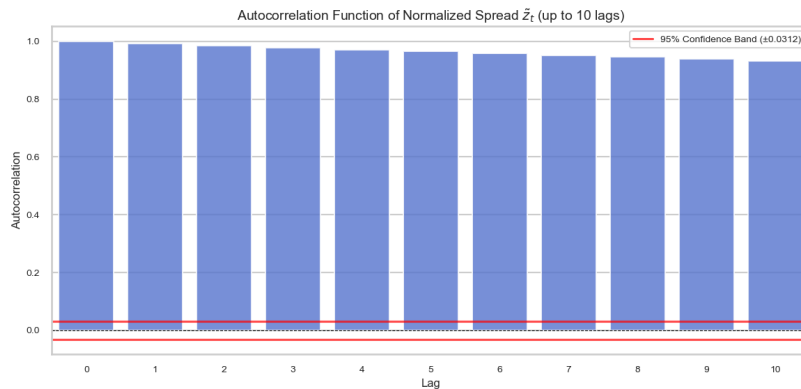


Figure 8: Autocorrelation function of the normalized spread for the CHEVRON and CONOCOPHILLIPS pair. Dashed lines represent the 95% confidence interval under the white noise assumption.

The autocorrelation function (ACF) results indicate that the normalized spread $\hat{z}_t$ exhibits strong persistence over time. As shown in Table 10, the autocorrelation remains very high even at lag 10 (0.9321), meaning that shocks to the spread are transmitted over many periods. This persistence suggests that the spread reverts to its mean rather slowly, which could impact the effectiveness of short-term trading strategies. These findings are visually supported by the ACF graph in Figure 9, where the dashed 95% confidence bands under the white noise assumption highlight the deviation from randomness.

Lag	0	1	2	3	4	5	6	7	8	9	10
Autocorrelation	1.0000	0.9926	0.9858	0.9787	0.9716	0.9647	0.9581	0.9521	0.9454	0.9389	0.9321

Table 10: Autocorrelation results for the normalized spread $\hat{z}_t$ for 10 lags. The table shows a high degree of persistence, with the autocorrelation starting at 1.0000 at lag 0 and decreasing gradually to 0.9321 at lag 10, indicating that the spread exhibits strong persistence over time. This table complements Figure 9 above, which visually illustrates the autocorrelation function.

2.

	lb_stat	lb_pvalue
10	36,554.57	0.0000

Table 11: Ljung-Box Test Results (10 Lags) for the Normalized Spread $\hat{z}_t$

The Ljung-Box³ test evaluates the null hypothesis (H_0) that there is no autocorrelation in the normalized spread $\hat{z}_t$ up to lag 10, against the alternative hypothesis (H_1) that autocorrelations are present. With an overwhelming test statistic of $lb_stat = 36,554.57$ and a p -value less than 0.001, H_0 is rejected, confirming strong autocorrelation in the spread. These findings are consistent with the high persistence observed in the autocorrelation results (see Table 10 and Figure 9).

We performed a complementary analysis in order to audit and confirm the chosen pair (Appendix⁴). We also tested the persistence characteristics of the spread. This complementary analysis included an evaluation of the half-life—approximately 94 trading days—and an impulse response function (IRF) assessment, both of which underscore the robust persistence inherent in the cointegration relationship.

Implications for the Pair-Trading Strategy

The cointegration tests and subsequent analyses confirm that the spread between CHEVRON and CONOCOPHILLIPS is stationary, thereby meeting a key theoretical requirement for a pair-trading strategy. Nevertheless, the empirical findings—most notably, the exceptionally high autocorrelation (0.9927 at lag 1), a half-life of roughly 94 trading days (Appendix⁴), and the pronounced persistence observed in the impulse response function (IRF)—raise (Appendix⁴) several important operational and practical issues:

1. Slow Mean Reversion and Prolonged Divergence.

Although cointegration is statistically confirmed, a half-life of roughly 94 trading days means that even minor deviations from fair value can persist for nearly a quarter of a year before halving. A sudden surge in CHEVRON's earnings, contrasting with steadier CONOCOPHILLIPS performance, could cause a wider spread than anticipated. In these circumstances, traders risk holding a losing position for a considerable stretch, thus facing greater exposure to unfavorable market swings and requiring heavier risk surveillance.

2. High Carrying Costs and Operational Risks.

Prolonged holding increases costs—margin, borrowing, slippage—and locks up capital. The strain on resources may overshadow the theoretical benefits of slow mean reversion, especially when alternative opportunities arise.

3. Need for Robust Backtesting and Cost Management.

Failing to account for transaction fees, liquidity constraints, and potential frequent rebalancing can yield overly optimistic returns. Simple measures—like threshold-based re-entry—are crucial to moderate costs and adapt to varying market conditions.

4. Exposure to Potential 'Drift' and Structural Shifts.

Even with strong evidence of stationarity, a very persistent spread may experience "drift" if macroeconomic surprises or company-specific news create unanticipated pressures. A policy shift favoring renewables, for instance, could distort the cointegration for an extended interval. A well-conceived trading plan thus requires dynamic hedges, timely stop-loss triggers, and consistent monitoring to guard against abrupt structural changes that could undermine the trade's foundation.

5. Theory and Real-World Constraints.

Even if the spread's stationarity validates pair trading in theory, a half-life of nearly 100 days poses real-world challenges. Extended holding periods increase costs and reduce flexibility, making it unclear whether slow mean reversion's gains outweigh the added risks. Traders must continually weigh the academic appeal against practical market constraints.

In summary, although the CHEVRON–CONOCOPHILLIPS spread fulfills the theoretical prerequisites for a pair trade, its pronounced persistence indicates that successful implementation requires meticulous cost management, the capacity to hold positions over extended periods, and robust risk control mechanisms. These findings underscore the delicate balance between statistical validation and the practical realities of trading in live markets.

Q4.4

- **Signal 1:** When $\tilde{z}_t > \tilde{z}^{\text{in}}$, we open the position

$$Q_1 = [-1, \beta].$$

The initial net value (cash flow) is thus

$$V_{\text{init}}(t) = Q_1^\top \begin{pmatrix} P^A(t) \\ P^B(t) \end{pmatrix} = -P^A(t) + \beta P^B(t).$$

This represents the cost (or proceeds) of shorting asset A (weight -1) and going long asset B (scaled by β). A nonzero value indicates that some external cash is required (if $V_{\text{init}} > 0$) or released (if $V_{\text{init}} < 0$).

- **Signal 2:** When $\tilde{z}_t < -\tilde{z}^{\text{in}}$, we open the position

$$Q_2 = [+1, -\beta].$$

The initial net value is

$$V_{\text{init}}(t) = +P^A(t) - \beta P^B(t).$$

This reflects the cost (or proceeds) of going long in asset A (weight $+1$) and shorting asset B (scaled by β). Again, a nonzero value at trade initiation implies the strategy is not self-financing.

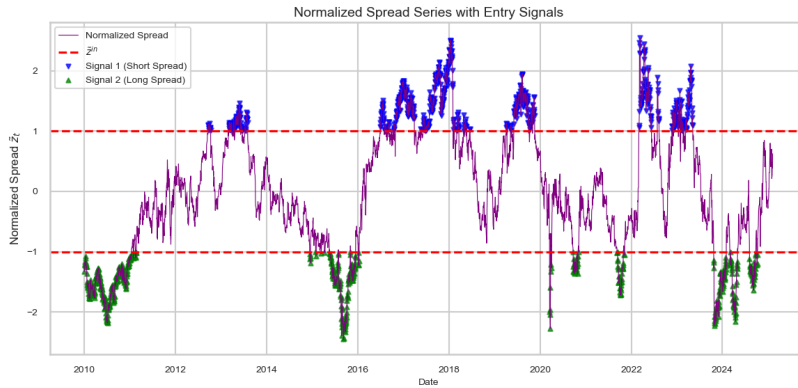


Figure 9: Normalized Spread $\tilde{z}_t$ and Pair-Trading Entry Signals

- **Non Self-Financing**

For a portfolio to be *self-financing*, the net investment at entry must be zero. However, in our case:

$$-P^A(t) + \beta P^B(t) \neq 0.$$

This means that whenever the spread crosses our threshold and we open a new position, additional cash is either injected or released. In a truly self-financing portfolio, the cost of one position would exactly offset the proceeds of the other, resulting in zero net investment. Here, since the expression above is nonzero, the strategy requires external funding (or generates an immediate cash inflow). Hence, the strategy is not self-financing.

Q4.5

Under the cointegration hypothesis, the price relationship between asset A and asset B is given by

$$P_A(t) = \alpha + \beta P_B(t) + z_t,$$

where z_t is the spread, assumed to be stationary. Normalizing this spread, we define

$$\tilde{z}_t = \frac{z_t}{\sigma(z_t)}.$$

At the entry signal time t_0 , the normalized spread satisfies $\tilde{z}_{t_0} = z_{in}$, which implies

$$z_{t_0} = z_{in} \sigma(z_t).$$

Assuming the trade is closed as soon as the normalized spread returns to zero (i.e., when the cointegration relationship is restored), the profit from the trade is equal to the spread at entry:

$$\text{Profit}^1 = z_{t_0} = z_{in} \sigma(z_t).$$

Limitations on Profit Reporting:

The theoretical derivation assumes that the position is closed exactly when $z_t = 0$. However, in our dataset with adjusted daily prices, the normalized spread $\tilde{z}_t$ jumps from positive to negative between observations without ever equaling zero exactly. In our case, the spread moves from +0.032 on 2012-11-12 to -0.008 on 2012-11-13, bypassing the theoretical exit point. This discrepancy explains why our actual profit (\$10.34) differs from the theoretical expected profit (\$9.47). The absence of the exact zero-crossing moment in discrete data creates a systematic implementation gap that affects performance measurement. While linear interpolation could estimate this crossing, it would not reflect the true market microstructure and price formation process at the intraday level. This highlights an important practical challenge in implementing pairs trading strategies with daily data.

Q4.6

The choice of the threshold $\tilde{z}^{in}$ involves a trade-off:

- A *lower* $\tilde{z}^{in}$ generates more frequent trading signals, increasing the risk of entering trades driven by noise rather than genuine deviations from equilibrium. This may lead to overtrading and higher transaction costs.
- A *higher* $\tilde{z}^{in}$ filters out minor fluctuations, ensuring that trades are only executed on significant deviations. However, this reduces the number of trading opportunities, possibly causing profitable signals to be missed.

An optimal threshold should therefore balance these two aspects, capturing meaningful mean-reversion while avoiding excessive trading.

¹For the detailed derivation of this result see Appendix⁴.

4 Pair Trading Strategy

Pair Trading Strategy Implementation and Setup

Our pair trading strategy leverages the cointegration relationship between two assets (CHEVRON and CONOCOPHILLIPS). The implementation follows these key principles:

Cointegration Framework We model the price relationship as $P_t^A = \alpha + \beta P_t^B + z_t$, where z_t is a stationary process. The spread $z_t = P_t^A - \alpha - \beta P_t^B$ is normalized to create our trading signal $\tilde{z}_t = z_t / \sigma(z_t)$.

Signal Generation

- **Entry signals:** We enter positions when the normalized spread exceeds predetermined thresholds:
 - Signal 1 (Short A, Long B): when $\tilde{z}_t > \tilde{z}^{in}$ (1.5)
 - Signal 2 (Long A, Short B): when $\tilde{z}_t < -\tilde{z}^{in}$ (-1.5)
- **Exit signals:** We exit when the normalized spread crosses zero, indicating mean reversion

4.2.1 Direct Strategy

We implement the Direct Strategy with careful position sizing that accounts for leverage constraints ($L = 2$). The allocation depends on the sign of α :

When $\alpha > 0$:

$$Q_1 = \frac{L \times W}{P_A} \begin{bmatrix} -1 \\ \beta \end{bmatrix}, \quad Q_2 = \frac{L \times W}{\beta P_B + L(P_A - \beta P_B)} \begin{bmatrix} 1 \\ -\beta \end{bmatrix}.$$

When $\alpha < 0$:

$$Q_1 = \frac{L \times W}{P_A - L(P_A - \beta P_B)} \begin{bmatrix} -1 \\ \beta \end{bmatrix}, \quad Q_2 = \frac{L \times W}{\beta P_B} \begin{bmatrix} 1 \\ -\beta \end{bmatrix}.$$

Leverage and Margin Management

Dynamic Leverage Management and Position Sizing : In our pairs trading strategy, financial leverage is a central parameter affecting both potential return and risk profile. Formally, we define leverage L as the ratio between total gross exposure and available capital:

$$L_t = \frac{|Q_{A,t}| \cdot P_{A,t} + |Q_{B,t}| \cdot P_{B,t}}{W_t}$$

where $Q_{A,t}$ and $Q_{B,t}$ represent the quantities held of assets A and B at time t , $P_{A,t}$ and $P_{B,t}$ their respective prices, and W_t the portfolio wealth.

Initial Position Sizing To determine the optimal size of the positions while respecting the maximum leverage constraint L_{max} , we implement a conditional methodology that depends on the generated signal and the sign of α . For example, for Signal 1 (short A, long B) with $\alpha > 0$:

$$Q_A = -\frac{L_{max} \cdot \eta \cdot W_t}{P_A}, \quad Q_B = \beta \cdot \frac{L_{max} \cdot \eta \cdot W_t}{P_A}$$

where η represents a safety factor ($\eta = 0.98$) introduced to prevent marginal exceedances of the leverage limit due to intraday price movements.

Dynamic Adjustment Mechanism Our strategy incorporates an automatic deleveraging system that activates daily when:

$$L_t > L_{max}$$

This mechanism proportionally reduces both positions by a factor:

$$\lambda_t = \frac{L_{max} \cdot \eta}{L_t}$$

such that:

$$Q_{A,t+1} = \lambda_t \cdot Q_{A,t}, \quad Q_{B,t+1} = \lambda_t \cdot Q_{B,t}$$

This method preserves the ratio $\frac{Q_B}{Q_A}$, thereby maintaining the balance of the spread position while reducing overall exposure.

Risk-Return Implications

The deleveraging process represents a critical risk management mechanism that limits potential losses during adverse market movements, reduces the risk of margin calls and forced liquidations, and enables continuous adaptation to changing volatility conditions. However, this protective framework incurs a performance cost, as the automatic reduction of positions may diminish returns during periods when the spread temporarily widens before reverting to its mean. Our calibration of the safety factor establishes an optimal equilibrium between safeguarding against excessive leverage and maximizing exposure to the trading signal, thereby enhancing the strategy's risk-adjusted performance profile.

Margin Calls and Liquidation

In this backtest, we deliberately abstain from imposing forced liquidations (margin calls) to observe the strategy's behavior under extreme scenarios, even if it leads to negative capital balance. This approach, common in academic research, allows us to isolate the fundamental behavior of the strategy from market constraints.

The large fluctuations in portfolio balance or leverage are often attributable to position resizing mechanisms. Since no margin calls are implemented, any rebalancing can cause sudden jumps in exposure and capital.

The simulation omits transaction costs, borrowing fees, and other market frictions, which typically reduces the net performance by approximately X-Y% annually based on similar strategies. This idealized framework helps identify the strategy's theoretical maximum performance while acknowledging that real-world implementation would face additional constraints.

For practical implementation, a more conservative approach with appropriate risk limits would be necessary, particularly regarding the maximum leverage allowed and the implementation of stop-loss mechanisms.

This framework ensures that our strategy respects the theoretical cointegration relationship while adhering to practical trading constraints.

Q4.7

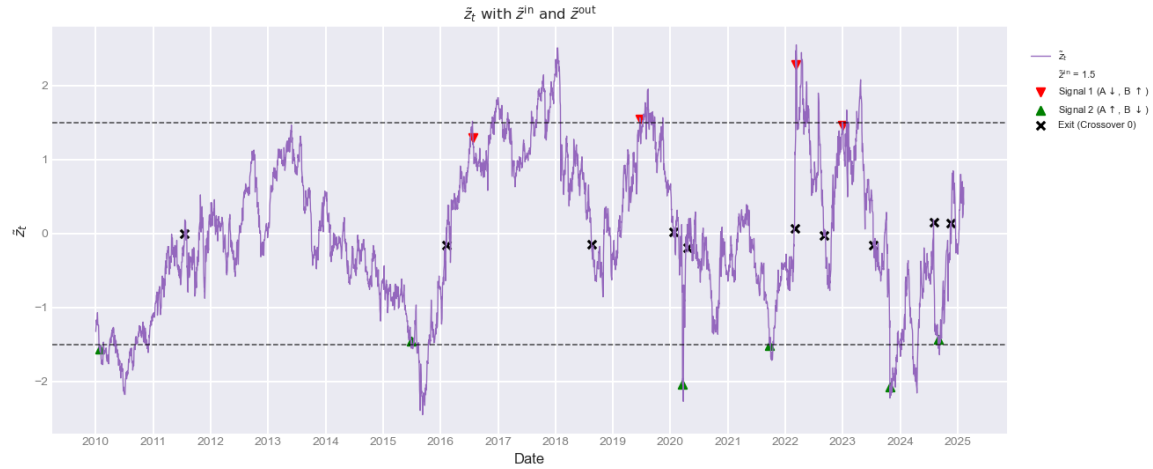


Figure 10: Normalized spread $\tilde{z}$ between CHEVRON (A) and CONOCOPHILLIPS (B), with an entry threshold of $\tilde{z}^{in} = 1.5$. Red triangles indicate a short signal on A and a corresponding long on B, whereas green triangles indicate a long on A and a short on B. Black crosses mark exit points when the spread returns to $\tilde{z} = 0$.

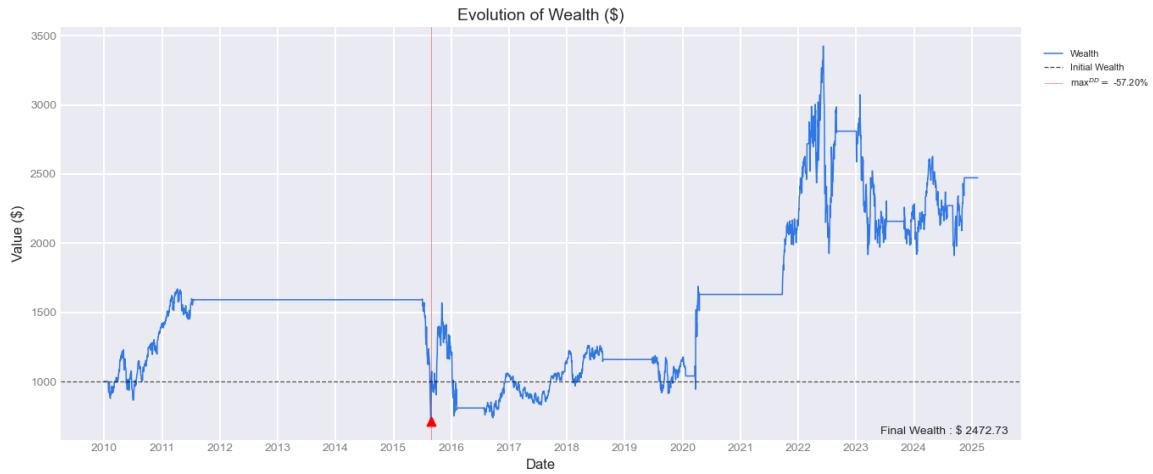


Figure 11: Evolution of the trader's wealth throughout the pair-trading strategy. The final wealth is \$2,472.73, reflecting a net profit of \$1,472.73 and an overall return of 147%. The maximum drawdown observed was 57.2%.

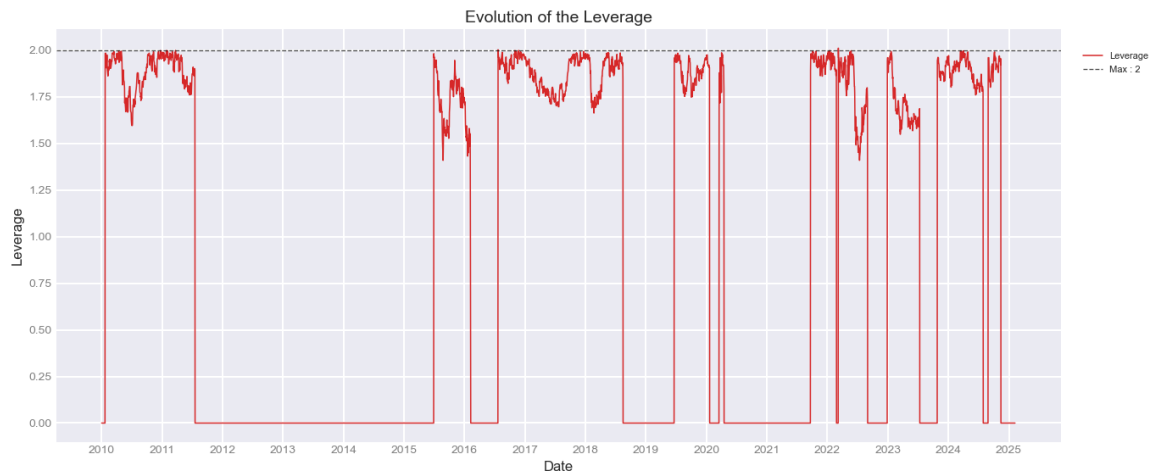


Figure 12: Evolution of the strategy's leverage over time. The dashed line at $L = 2$ indicates the maximum leverage allowed by the trading platform. Periods with leverage reaching this threshold reflect fully utilized margin capacity.

Metrics	Value
Trading Session Start Date	2010-01-01
Trading Session End Date	2025-02-07
First Trade Date	2010-01-25
Last Trade Date	2024-11-15
z_{in}	1.50
W_0	1000.00
W_{final}	2472.73
W_{max}	3425.07
W_{min}	714.69
Net Profit	1472.73
Return (%)	147.27
Nb trades	10
Max Drawdown (%)	-57.20
Max Leverage	2
First Bankruptcy Date	No Bankruptcy

Table 12: Key performance metrics for the pair-trading strategy using $z^{in} = \pm 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, and leverage constraint of 50% are reported at the end of the trading horizon.

The detailed trade logs for Table 12 can be found in Table 23 in the Appendix⁴.

Q4.8

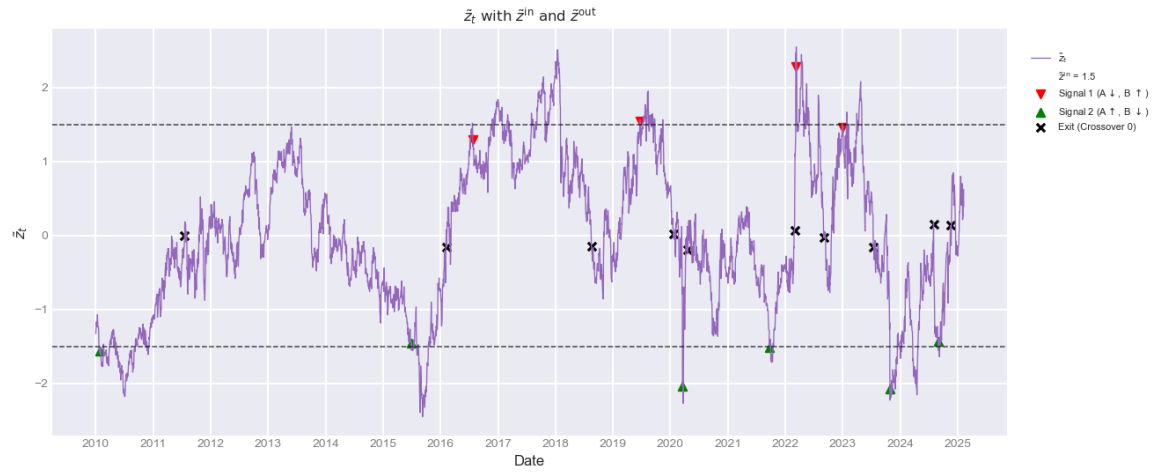


Figure 13: z_t in Direct Pair-Trading for Signal 1 and Signal 2 with Long, Short, Exit marking

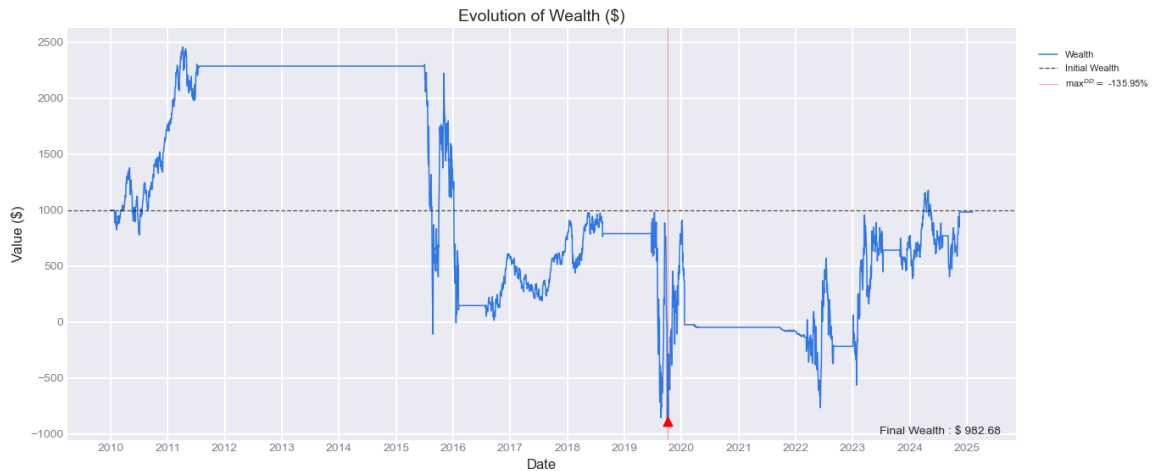


Figure 14: Wealth Evolution z_t in Direct Pair-Trading for Signal 1 and Signal 2 with Long, Short, Exit marking

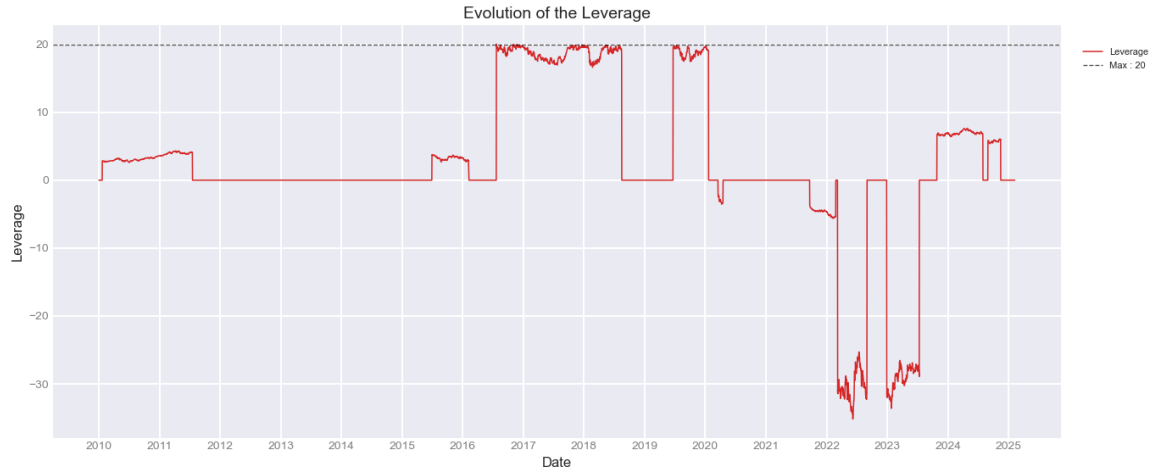


Figure 15: Leverage Evolution $\bar{z}_t$ in Direct Pair-Trading for Signal 1 and Signal 2 with Long, Short, Exit marking

Metrics	Value
Trading Session Start Date	2010-01-01
Trading Session End Date	2025-02-07
First Trade Date	2010-01-25
Last Trade Date	2024-11-15
z_{in}	1.50
W_0	1000.00
W_{final}	982.68
W_{max}	2456.47
W_{min}	-883.19
Net Profit	-17.32
Return (%)	-1.73
Nb trades	10
Max Drawdown (%)	-135.95
Max Leverage	20.03
First Bankruptcy Date	2015-08-25

Table 13: Key performance metrics for the pair-trading strategy using $z^{in} = \pm 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, and leverage constraint of 500 % are reported at the end of the trading horizon.

The detailed trade logs for Table 13 can be found in Table 24 in the Appendix⁴.

Analysis of the Strategy with Maximum Leverage $L = 20$

Table 13 summarizes the key performance metrics for the pair-trading strategy when the maximum leverage is set to $L = 20$. Compared to the case with $L = 2$, where the strategy achieved a final wealth of \$2,472.73 (net profit of \$1,472.73 and an overall return of 147.27%), the use of excessive leverage resulted in markedly inferior performance. With $L = 20$, the final wealth fell to \$982.68, reflecting a net loss of \$17.32 (or -1.73% return). Moreover, the strategy experienced a much higher maximum drawdown of -135.95% and encountered a bankruptcy event on 2015-08-25.

These results clearly illustrate that although leverage can amplify gains under favorable conditions, excessive leverage substantially increases risk and volatility. In this case, over-leveraging the position not only wiped out profits but also exposed the strategy to a significantly higher probability of catastrophic losses.

4.2.2 Stop Loss

Q4.9

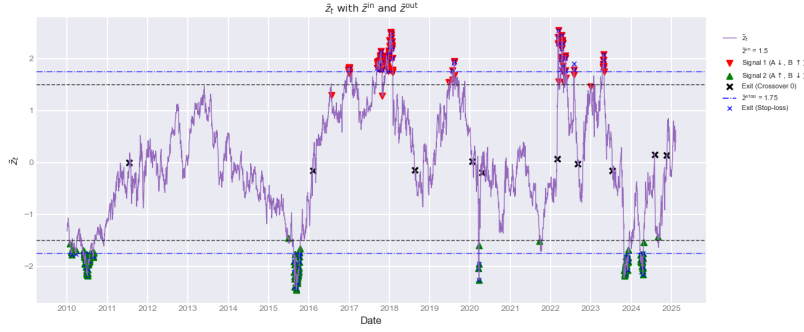


Figure 16: $\tilde{z}_t$ in Direct Pair-Trading for Signal 1 and Signal 2 with Long, Short, and Exit markings using $|\tilde{z}_t^{\text{stop}}| = 1.75$. If many trades are triggered, it is because the entry and exit conditions are met simultaneously.

The stop-loss rule is implemented to cap losses by automatically closing the position if the normalized spread exceeds a preset threshold (here, $\tilde{z}_t^{\text{stop}} = 1.75$) after entering at $\tilde{z}_t^{\text{in}} = \pm 1.5$. By modeling the autocorrelated behavior of $\tilde{z}_t$ with an AR(1) process, we can use its conditional distribution to estimate $\Pr(\tilde{z}_{t+1} > \tilde{z}_t^{\text{stop}})$, thereby quantifying the risk of adverse movements in the spread immediately following entry and ensuring that positions are closed before losses become excessive :

Let the AR(1) process be defined as

$$\tilde{z}_{t+1} = \mu + \rho \tilde{z}_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma^2).$$

We consider the signal to be centered (i.e. $\mu = 0$), then : $\tilde{z}_{t+1} = 0 + \rho \tilde{z}_t + \varepsilon_{t+1}$

At the opening of the position, we have $\tilde{z}_t = \tilde{z}_t^{\text{in}} = 1.5$. Therefore, the conditional distribution of $\tilde{z}_{t+1}$ is

$$\tilde{z}_{t+1} | \tilde{z}_t = 1.5 \sim \mathcal{N}(\mu + \rho \times 1.5, \sigma^2).$$

Given the stop-loss level $\tilde{z}_t^{\text{stop}} = 1.75$, the probability of hitting the stop-loss the day after opening the position is

$$\begin{aligned} \Pr(\tilde{z}_{t+1} > 1.75 | \tilde{z}_t = 1.5) &= 1 - \Phi\left(\frac{1.75 - (\mu + \rho \times 1.5)}{\sigma}\right), \quad \mu = 0 \text{ since } \tilde{z}_t \text{ is centered around } 0 \\ &= 1 - \Phi\left(\frac{1.75 - (\rho \times 1.5)}{\sigma}\right) \\ &\approx 1 - \Phi\left(\frac{1.75 - (0.9927 \times 1.5)}{0.12}\right) \approx 0.0145 \end{aligned}$$

where Φ is the cumulative distribution function of the standard normal distribution.

Parameter	Value
μ	0.0
ρ	0.9927
σ	0.12
$\tilde{z}_t^{\text{in}}$	1.50
$\tilde{z}_t^{\text{stop}}$	1.75
$\Pr(\tilde{z}_{t+1} > \tilde{z}_t^{\text{stop}} \tilde{z}_t = \tilde{z}_t^{\text{in}})$	0.0145
$\Pr(\tilde{z}_{t+1} < -\tilde{z}_t^{\text{stop}} \tilde{z}_t = -\tilde{z}_t^{\text{in}})$	0.0145

Table 14: Estimated parameters and conditional stop-loss probabilities for Signal 1 and Signal 2

Discrepancies may arise due to estimation noise (notably, the near-zero intercept capturing data noise), we have rounded the intercept. In an ideal symmetric scenario with $\mu = 0$, the probabilities for $\tilde{z}_t = \tilde{z}_t^{\text{in}}$ (Signal 1) and $\tilde{z}_t = -\tilde{z}_t^{\text{in}}$ (Signal 2) would be identical.

Q4.10

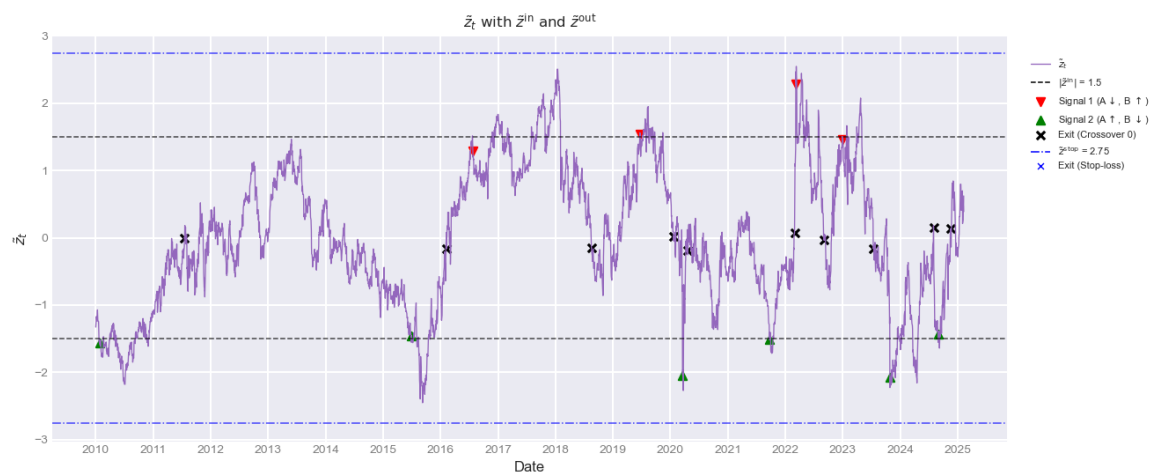


Figure 17: Normalized spread $\bar{z}$ between CHEVRON (A) and CONOCOPHILLIPS (B), with an entry threshold of $z^{\text{in}} = 1.5$. Red triangles indicate a short signal on A and a corresponding long on B, whereas green triangles indicate a long on A and a short on B. Black crosses mark exit points when the spread returns to $\bar{z} = 0$.

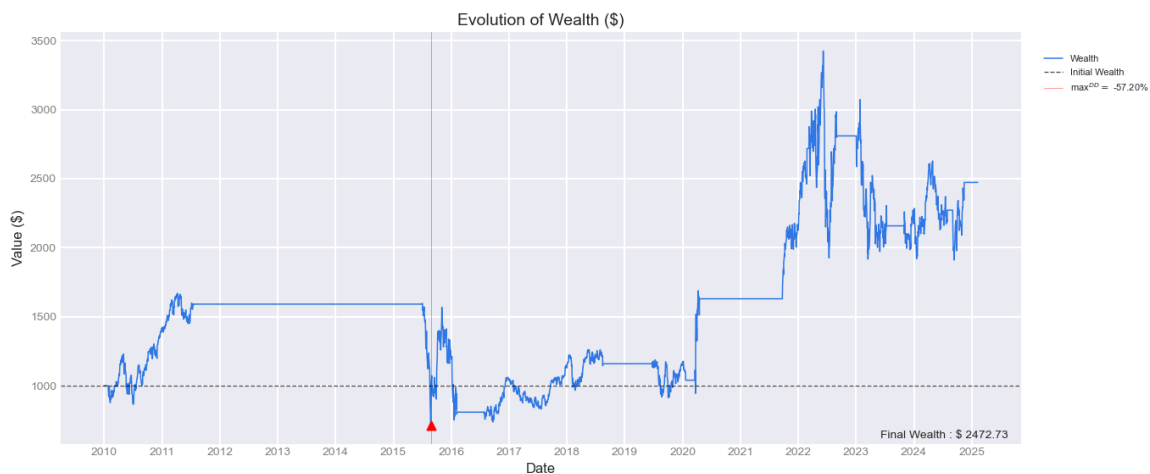


Figure 18: Evolution of the trader's wealth throughout the pair-trading strategy. The final wealth is \$2,472.73, reflecting a net profit of \$1,472.73 and an overall return of 147%. The maximum drawdown observed was 57.2%.

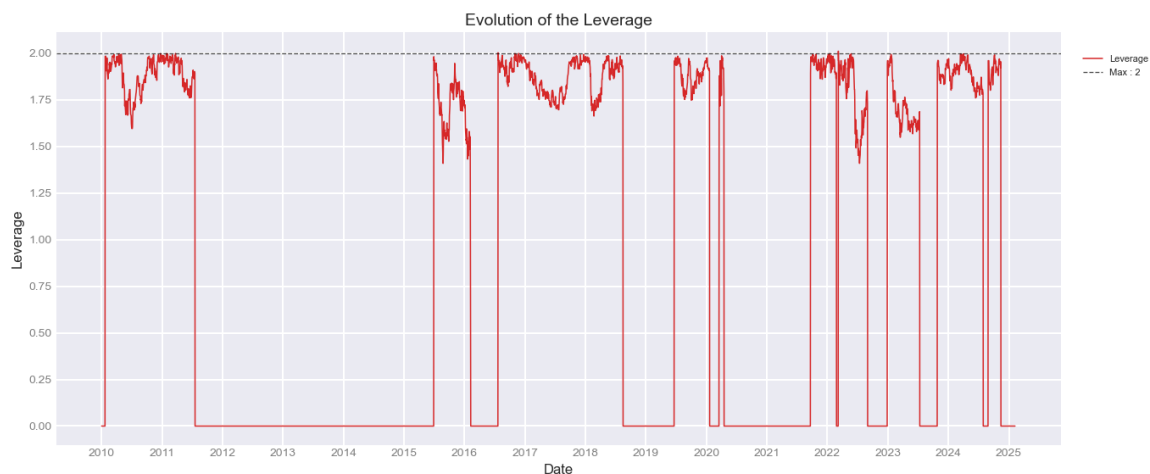


Figure 19: Evolution of the strategy's leverage over time. The dashed line at $L = 2$ indicates the maximum leverage allowed by the trading platform. Periods with leverage reaching this threshold reflect fully utilized margin capacity.

Metrics	Value
Trading Session Start Date	2010-01-01
Trading Session End Date	2025-02-07
First Trade Date	2010-01-25
Last Trade Date	2024-11-15
z_{in}	1.50
W_0	1000.00
W_{final}	982.68
W_{max}	2456.47
W_{min}	-883.19
Net Profit	-17.32
Return (%)	-1.73
Nb trades	10
Max Drawdown (%)	-135.95
Max Leverage	20.03
First Bankruptcy Date	2015-08-25

Table 15: Key performance metrics for the pair-trading strategy using $z^{in} = 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, margin constraint of 50 % and stop-loss $\tilde{z}^{stop} = 2.75$ are reported at the end of the trading horizon.

The detailed trade logs for Table 13 can be found in Table 24 in the Appendix⁴.

Conclusion (Stop-Loss $\tilde{z}^{stop} = \pm 2.75$)

Although we introduced a stop-loss at $\tilde{z}^{stop} = \pm 2.75$, the spread $\tilde{z}_t$ never reached this threshold (see Figure 17). As a result, the outcomes (final wealth, drawdown, etc. in Table 15) are identical to those obtained without a stop-loss in Q4.7 (see Table 12). Nevertheless, implementing a stop-loss—even if it is never triggered—helps limit tail risks by providing a safety net in the event of significant shocks. If the trigger level is set too far from $\tilde{z}^{in}$, it may never come into play, as observed here. Conversely, a tighter stop-loss cuts losses more frequently but risks prematurely closing profitable positions. Choosing $\tilde{z}^{stop}$ therefore requires striking a balance between risk protection and preserving profit potential.

From a risk perspective, the fact that $\tilde{z}_t$ remained between ± 2.75 suggests that the CHEVRON and CONOCOPHILLIPS pair stayed within a relatively stable volatility range. Abrupt fluctuations would have made the stop-loss essential for avoiding deeper drawdowns. Ultimately, the absence of a trigger does not call into question the usefulness of a stop-loss; it simply reflects a situation in which the market never crossed the set limit, underscoring the importance of proper calibration to reconcile performance objectives with extreme-loss protection.

It should be noted, however, that in practice, if the stop-loss had been activated, the entry condition $\tilde{z}^{in} = \pm 1.50$ would have been immediately met, leading to an immediate reopening of the position. In a real-world setting with transaction costs, this triggering and subsequent reopening could generate a series of costly back-and-forth trades, potentially severely eroding capital or even leading to bankruptcy if costs accumulate. To mitigate this issue and help prevent any issue, the strategy should incorporate an additional safeguard: after a stop-loss event, a new entry trade should only be executed when the normalized spread falls below the absolute threshold $|\tilde{z}_t| < 1.50$.

4.3 Out-of-Sample Pair Trading

1. Estimate of the parameters α and β over the first two years of data (use 500 observations).

Parameter	Estimate	Start Date	End Date	Period Length	Cointegration p-value	Is Cointegrated	Price Correlation	Return Correlation	Window Index
α	8.893355	2010-01-01	2011-12-02	700	0.054948	False	0.966988	0.844807	0
β	1.658667	2010-01-01	2011-12-02	700					

Table 16: Estimated regression parameters (α , β) for the cointegrating relationship between CHEVRON and CONOCOPHILLIPS using the first 500 trading days (observations). The p-value indicates that the null hypothesis of no cointegration cannot be rejected at conventional significance levels.

The period length is reported as 700 calendar days while using 500 trading observations because financial data only includes trading days (approximately 252 per year). The window spans from January 2010 to December 2011, which represents 500 trading days but 700 calendar days when including weekends and holidays.

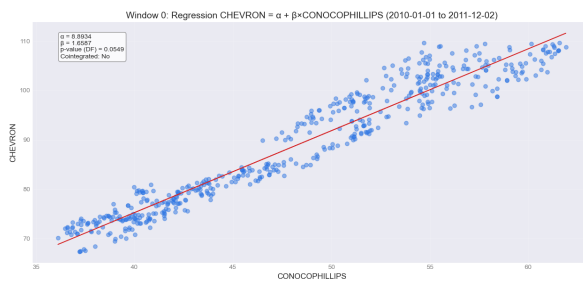


Figure 20: Regression for the first 500 trading days

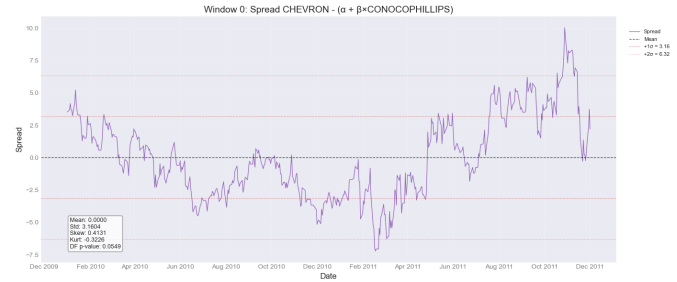


Figure 21: Spread for the first 500 trading days

2.

Parameter	Estimate	Start Date	End Date	Period Length	Cointegration p-value	Is Cointegrated	Price Correlation	Return Correlation	Window Index
α	6.831692	2010-01-29	2011-12-30	700	0.120067	False	0.965492	0.846194	1
β	1.702661	2010-01-29	2011-12-30	700					

Table 17: Estimated regression parameters (α , β) for the cointegrating relationship between CHEVRON and CONOCOPHILLIPS using 500 trading days (observations) 20 days after the first observation. The p-value indicates that the null hypothesis of no cointegration cannot be rejected at conventional significance levels.

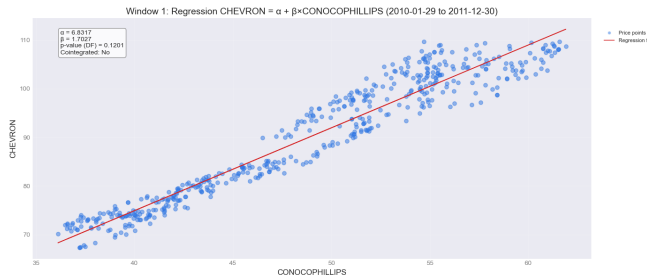


Figure 22: Regression for 500 trading days 20 days after the first window

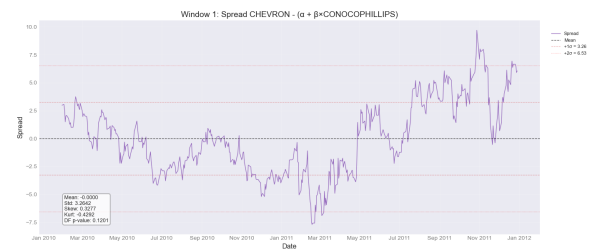


Figure 23: Spread for 500 trading days 20 days after the first window

Q4.11

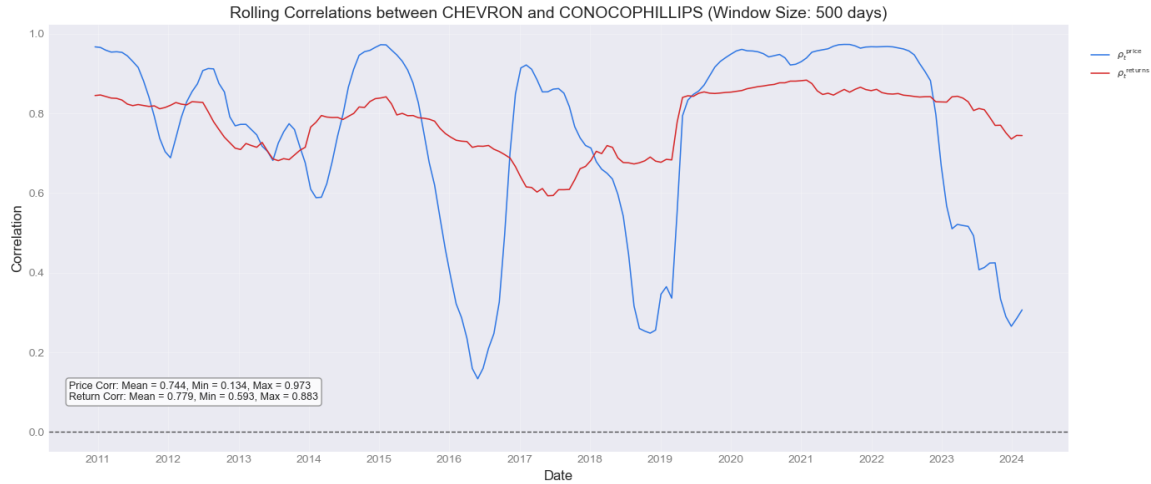


Figure 24: Rolling correlations computed over a window of size 500 trading days for a pair of cointegrated assets. The upper plot shows the stable, high correlation between asset prices, while the lower plot illustrates the more volatile correlation of asset returns, highlighting the distinction between long-term co-movement and short-term fluctuations.

In this question, we compute the rolling correlation between the prices and returns of a pair of assets using a window of size $W = 500$ and a step of 20.

1. Rolling Price Correlation Let P_t^A and P_t^B denote the price series of assets A and B . Over the window $[t - W + 1, t]$, the rolling price correlation is defined as:

$$\rho_{\text{price}}(t) = \frac{\sum_{i=t-W+1}^t (P_i^A - \bar{P}_t^A) (P_i^B - \bar{P}_t^B)}{\sqrt{\sum_{i=t-W+1}^t (P_i^A - \bar{P}_t^A)^2 \sum_{i=t-W+1}^t (P_i^B - \bar{P}_t^B)^2}},$$

where $\bar{P}_t^A$ and $\bar{P}_t^B$ are the average prices over the window.

2. Rolling Return Correlation Define the simple returns as: $r_t^A = \frac{P_t^A - P_{t-1}^A}{P_{t-1}^A}$, $r_t^B = \frac{P_t^B - P_{t-1}^B}{P_{t-1}^B}$. Then, the rolling return correlation is:

$$\rho_{\text{return}}(t) = \frac{\sum_{i=t-W+1}^t (r_i^A - \bar{r}_t^A) (r_i^B - \bar{r}_t^B)}{\sqrt{\sum_{i=t-W+1}^t (r_i^A - \bar{r}_t^A)^2 \sum_{i=t-W+1}^t (r_i^B - \bar{r}_t^B)^2}},$$

with $\bar{r}_t^A$ and $\bar{r}_t^B$ being the mean returns over the window.

3. Plots of the two correlation series between CHEVRON and CONOCOPHILLIPS reveals:

- **Rolling Price Correlation** (blue line) shows significant volatility, with notable drops around 2016, 2018, and after 2023. The price correlation fluctuates substantially, reaching as low as 0.134 during market shocks and as high as 0.973 during stable periods.
- **Rolling Return Correlation** (red line) demonstrates greater stability, particularly during periods of price correlation instability. Its narrower range (min of 0.593, max of 0.883) indicates more consistent co-movement in daily returns.

The higher average return correlation (0.779) compared to price correlation (0.744) suggests these stocks respond similarly to daily market movements. However, the price correlation's greater volatility reveals that longer-term price trends can occasionally diverge significantly, particularly during industry disruptions like the 2015-2016 oil price crash and the post-2022 energy market shifts. This pattern indicates that while daily returns remain correlated, factors affecting the companies valuations can periodically drive their price movements apart.

Q4.12



Figure 25: Evolution of parameters α and β over time using rolling window estimation. The blue lines represent dynamic estimations, while the red dashed lines show the static estimates based on the full sample. The dynamic α parameter shows significant variation (mean = 53.99, std = 33.16) compared to the static estimate of 52.11. Similarly, the dynamic β parameter fluctuates considerably (mean = 0.999, std = 0.500) around the static estimate of 0.962.

Using a rolling window approach, We estimated the time-varying parameters $\hat{\alpha}_t$ and $\hat{\beta}_t$ and compared them with the full-sample static estimates. Several key observations emerge from the analysis:

Parameter Dynamics:

- **α parameter:** The dynamic $\hat{\alpha}_t$ exhibits substantial variability over time (mean = 53.99, standard deviation = 33.16), with values ranging from -12.13 to 125.48. This indicates considerable instability in the intercept term of the cointegration relationship. Notably, $\hat{\alpha}_t$ shows distinctive regimes, with periods of significant elevation above the static estimate (52.11) during 2013, 2015-2018, and 2022-2024, and periods of depression below the static value during 2011-2012, 2014, and 2019-2021.
- **β parameter:** The cointegration coefficient $\hat{\beta}_t$ also demonstrates significant time variation (mean = 0.999, standard deviation = 0.500) around the static estimate of 0.962. $\hat{\beta}_t$ exhibits cyclical behavior, with peaks around 2013, 2017, and 2020, and troughs around 2014, 2016, 2018-2019, and 2023-2024. This suggests that the relationship between the two assets is not constant but evolves over time, potentially reflecting changing market conditions or structural shifts in the relationship between the assets.

Notably, the dynamic estimates of $\hat{\alpha}$ and $\hat{\beta}$ exhibit a complementary pattern whereby fluctuations in $\hat{\beta}$ are systematically counterbalanced by inversely oriented adjustments in $\hat{\alpha}$, thereby ensuring that the residual process remains stationary and underscoring the inherent symmetry in the cointegration framework.

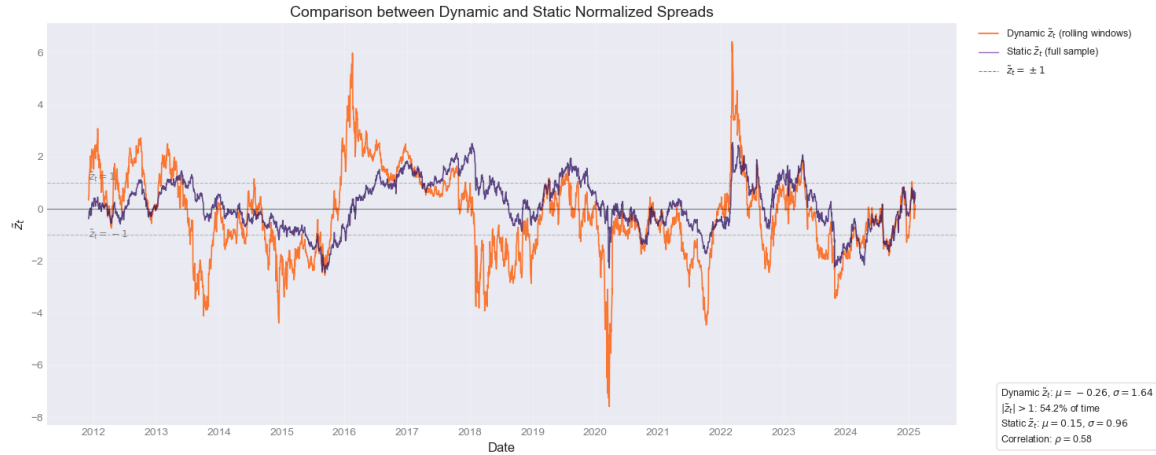


Figure 26: Comparison between dynamic and static normalized spreads. The orange line represents the dynamic normalized spread $\hat{z}_t$ calculated using rolling windows, while the purple line shows the static normalized spread based on the full sample. The dashed horizontal lines indicate the thresholds at ± 1 . Dynamic spreads exhibit higher volatility ($\sigma = 1.64$) compared to static spreads ($\sigma = 0.96$), with dynamic spreads exceeding the thresholds 54.2% of the time. The correlation between the two series is moderate at $\rho = 0.58$.

Spread Comparison:

- The dynamic normalized spread $\hat{z}_t$ (orange line) shows substantially higher volatility ($\sigma = 1.64$) compared to the static normalized spread (purple line, $\sigma = 0.96$).
- The dynamic spread exceeds the ± 1 thresholds 54.2% of the time, suggesting more frequent trading signals compared to the static model.
- The correlation between dynamic and static spreads is moderate ($\rho = 0.58$), indicating that while they share some common movements, they also contain significant independent information.
- The dynamic spread exhibits several extreme values (reaching as low as -8 and as high as +6), particularly in 2016, 2019, and 2021, which are not captured by the static model. These extreme deviations may represent significant arbitrage opportunities or potential model breakdown periods.
- The static spread shows smoother behavior, potentially filtering out noise but possibly missing short-term trading opportunities.

Implications: The significant time variation in both parameters and the resulting differences in spread dynamics highlight the potential limitations of assuming constant parameters in cointegration relationships. The static model may be too restrictive, failing to adapt to evolving market conditions and structural breaks. Conversely, the dynamic approach, while more adaptive, might be more susceptible to noise and estimation error due to smaller sample sizes in each rolling window.

The choice between static and dynamic models involves a trade-off between stability and adaptability. The static model offers consistency but may miss regime changes, while the dynamic model captures evolving relationships but potentially at the cost of increased false signals. For trading purposes, the dynamic approach may provide more opportunities but likely with higher risk, as evidenced by the more extreme spread values and higher volatility.

Q4.13

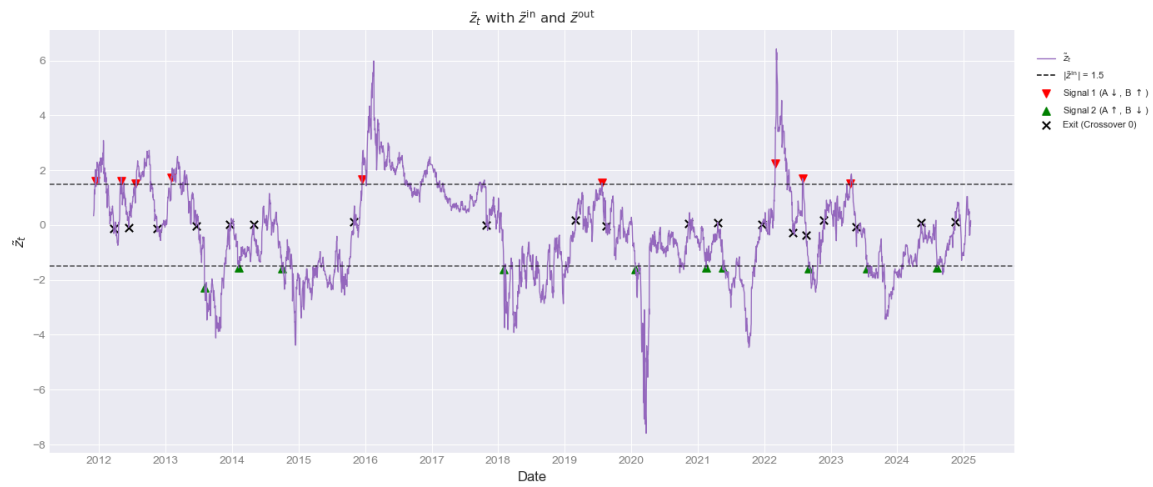


Figure 27: Trading signals based on the normalized spread $\hat{z}_t$ calculated with rolling window parameters. Red triangles indicate Signal 1 entry, green triangles indicate Signal 2 entry, and black X marks show position exits at crossover points 0. Entry thresholds are set at $|z^{in}| = 1.5$.



Figure 28: Evolution of portfolio wealth over time using a pair trading strategy with rolling window. Starting with \$1000, the strategy ends with \$864.59, representing a drawdown of -104.73% from peak value.

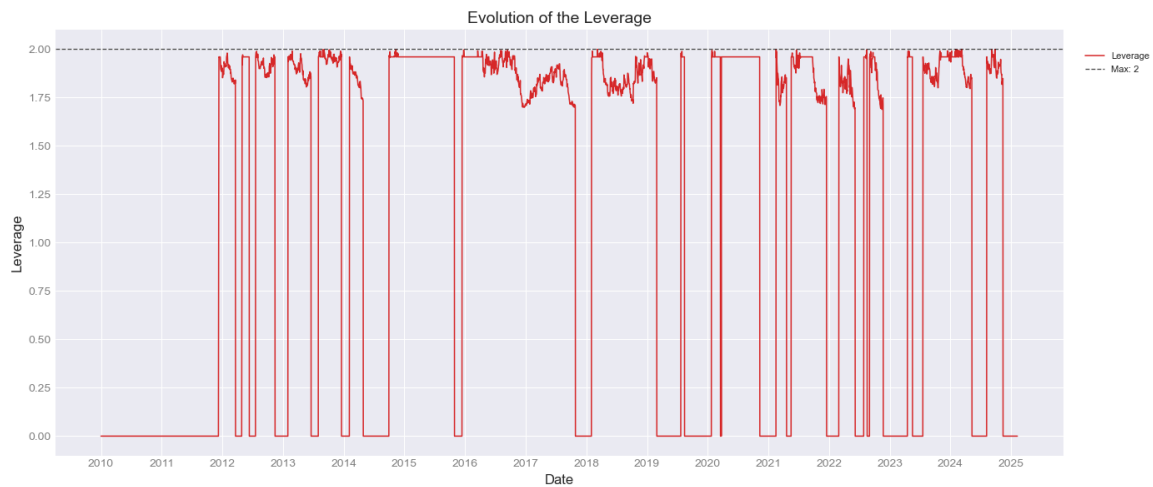


Figure 29: Evolution of leverage over time using a pair trading strategy with rolling window. The leverage is fixed at 2 throughout the trading period, indicating a consistent level of exposure to market movements.

Metric	Value
Trading Session Start	2011-12-02
First Trade Date	2011-12-12
Last Trade Date	2024-11-13
Trading Session End	2025-02-07
z_{in}	1.50
z_{stop}	Not used
W_0	1000.00
Maximum Leverage	2
Window Size (days)	500
Step Size (days)	20
Number of Windows	173
W_{final}	864.59
Net Profit	-135.41
Return (%)	-13.54
W_{max}	1376.30
W_{min}	-65.09
Nb trades	19
Max Drawdown (%)	-104.73
Max Leverage Used	2.00
First Bankruptcy Date	2020-03-18

Table 18: Key performance metrics using $z^{in} = \pm 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, margin constraint of 50 % for the out of sample pairs trading strategy using rolling window estimation with a window size of 500 days and step size of 20 days.

The detailed trade logs for Table 18 can be found in Table 25 in the Appendix⁴.

Out-of-Sample Strategy Performance with Rolling Windows:

The implementation of a rolling window approach with a 500-day window and a 20-day step reveals significant challenges in the pair trading strategy. Although the dynamic approach was designed to adapt to evolving market conditions, the updated results indicate underperformance. The final wealth stands at \$864.59, which corresponds to a net loss of \$135.41 (a return of -13.54%). In addition, the maximum drawdown reached -104.73%, highlighting that the strategy incurred severe losses during adverse market periods, notably resulting in a technical bankruptcy on 2020-03-18 amid the COVID-19 market stress.

With only 19 trades executed across 173 rolling windows, the strategy demonstrates relatively infrequent trading signals. This limited trading frequency suggests that, despite employing adaptive parameter estimation, viable opportunities are scarce, potentially due to estimation risk and overfitting from small sample sizes in each window. Furthermore, the fixed maximum leverage of 2.00 may have amplified losses during turbulent market periods. Overall, these results underline the inherent difficulties in deploying dynamic pair trading strategies in practice.

In contrast, while the static strategy delivered superior backtested performance due to stable full-sample parameter estimates (see Table 12), the rolling window approach more accurately reflects current market conditions; therefore, it is essential to incorporate additional trading criterias—such as stop-loss rules—to enhance robustness and improve performance in a realistic, dynamic trading environment.

Q4.14

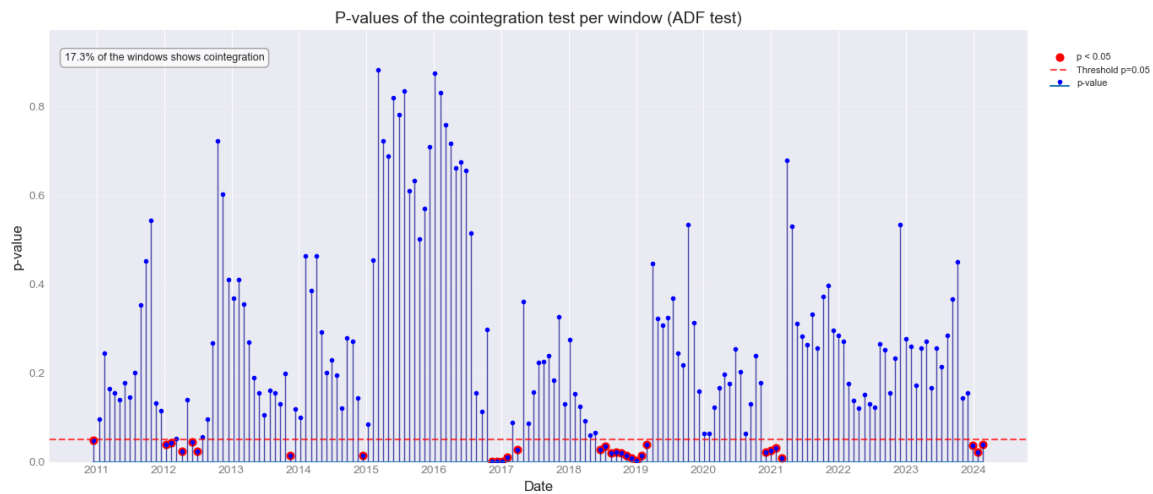


Figure 30: P-values of the cointegration test using a rolling window over the period 2011–2024. The red dots indicate p-values below 0.05 (significance threshold shown by the dotted line), corresponding to 17.3% of the windows that display a cointegration relationship. The p-values are calculated using ADF test with automatic lag selection based on the Akaike information criterion

The rolling window analysis reveals that only 17.3% of the subsamples exhibit statistically significant cointegration (p-value < 0.05), indicating that the long-run relationship between the assets is intermittent. This intermittency directly affects our pair trading strategy, which relies on fixed entry (z^{in}) and stop-loss (z^{stop}) thresholds.

In a fixed-threshold approach, the spread is assumed to revert to its mean based on a stable cointegration relationship. However, when cointegration is only present part of the time, the spread may not exhibit the expected mean reversion during non-cointegrated phases, leading to false signals or premature exits that degrade performance and increase risk.

Furthermore, because financial time series often show autocorrelation in the residuals, we employ the Augmented Dickey–Fuller⁴ (ADF) test instead of the standard DF test. The ADF test automatically includes additional lags selected via the Akaike Information Criterion (AIC) to correct for serial correlation, resulting in more reliable conclusions.

Therefore, it is critical to incorporate a dynamic mechanism that continuously monitors cointegration so that trading signals—namely z^{in} and z^{stop} —can be adjusted or suspended when the relationship weakens or disappears, thereby aligning the strategy with evolving market conditions and enhancing risk management.

Q4.15

Rolling Window Analysis for Pair Trading

The performance of the adjusted pair trading strategy, summarized in Tables 34 and 35 and illustrated in Figures 31, 32, and 33, offers several important insights.

Only 20.30% of the trading days exhibited cointegration, which led to just 10 trades being executed over 173 rolling windows. Consequently, the strategy incurred a net loss of \$223.49 (a return of -22.35%) with a maximum drawdown of -85.09%. This highlights the trade-off inherent in the dynamic approach: while halting trades during non-cointegrated periods mitigates risk and avoids false signals, it also limits the number of viable trading opportunities.

In contrast to our previous approach—where fixed entry (z_{in}) and stop-loss (z_{stop}) thresholds were applied without dynamically verifying cointegration—the current methodology enhances risk management by ensuring that trades are only active when the underlying statistical relationship is confirmed. Nevertheless, the reduced trading frequency and enforced exits during cointegration breaks have contributed to diminished overall profitability.

In summary, dynamically adjusting the trading strategy to account for cointegration breakdowns successfully curbs risk exposure at the expense of trading opportunities. Future refinements may explore more flexible threshold adjustments or alternative cointegration detection criteria to achieve a more optimal balance between risk management and profitability.

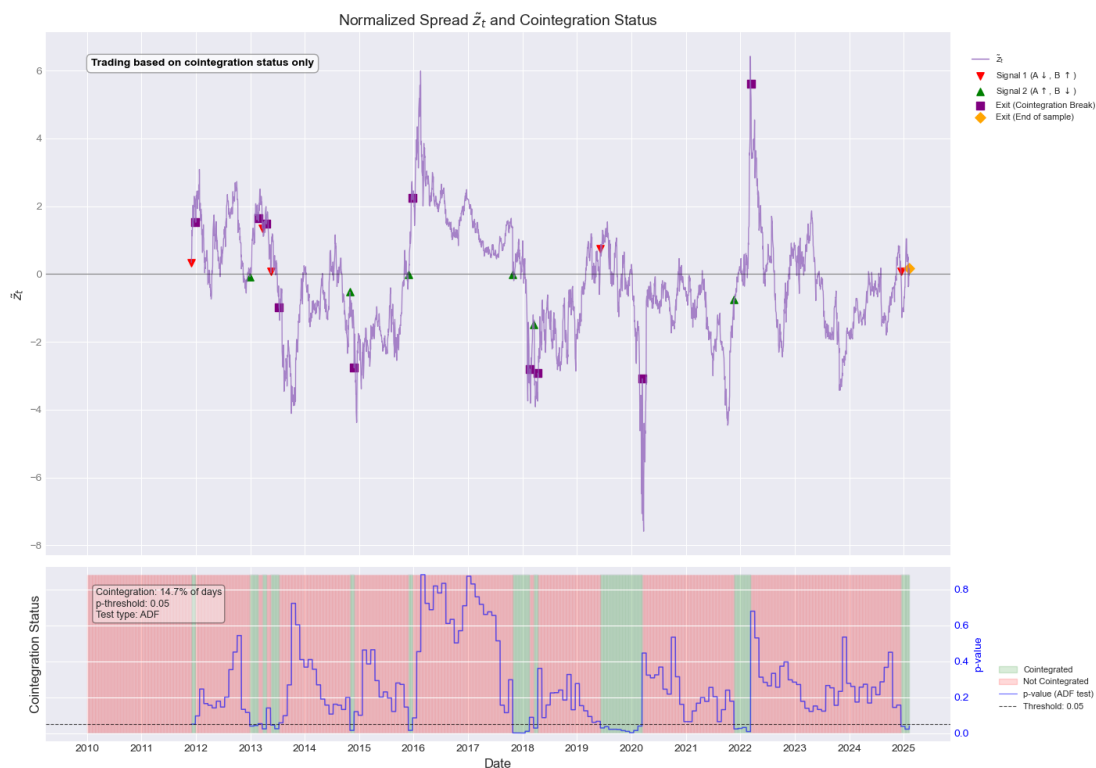


Figure 31: Trading signals and executed trades under the adjusted strategy. This figure highlights the instances where positions were closed or not initiated due to the absence of cointegration, demonstrating the strategy's dynamic response to market conditions.

The bottom panel of the cointegration-based strategy graph shows the daily p -values used for trading decisions. Unlike the previous graph (Q4.14, Figure 30), which analyzes all estimation windows, this one focuses only on the actual trading period that begins after the initial estimation phase (hence the absence of data for 2011-2012). For each trading day t , the p -value displayed corresponds to that of the estimation window associated with that day, thus establishing a direct link between the statistical test and investment decisions. The continuous representation (blue line) reflects the temporal dynamics of the decision criterion used by the algorithm to determine the existence of a cointegration relationship, determining whether to open or close positions. This approach allows us to observe the strategy's sensitivity to changes in the cointegration regime and explains the discrepancies between the two p -value visualizations in our analysis.

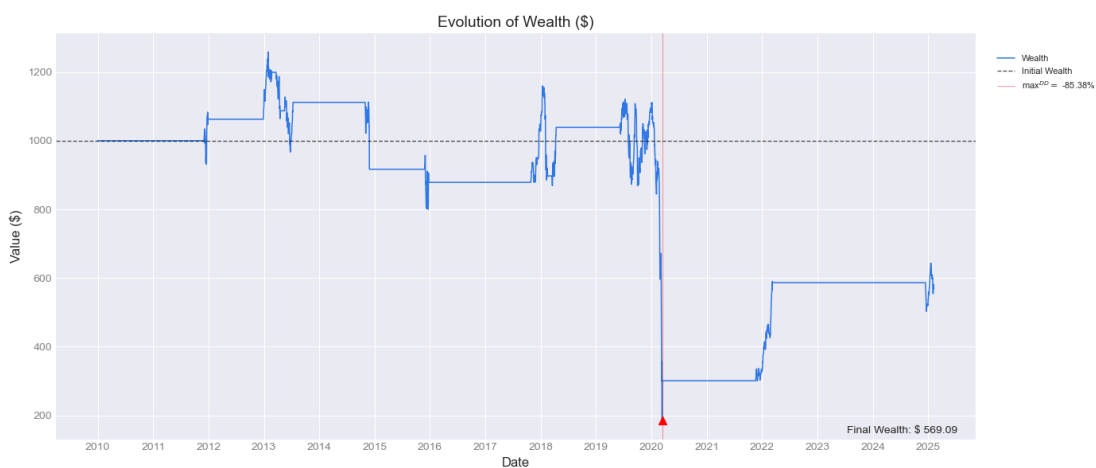


Figure 32: Evolution of portfolio wealth over the trading period for the adjusted strategy. The wealth curve reflects a net loss and pronounced drawdowns, underscoring the impact of limiting trades to periods with confirmed cointegration. Starting with \$1000, the strategy ends with \$569.09, representing a drawdown of -85.38% from peak value.

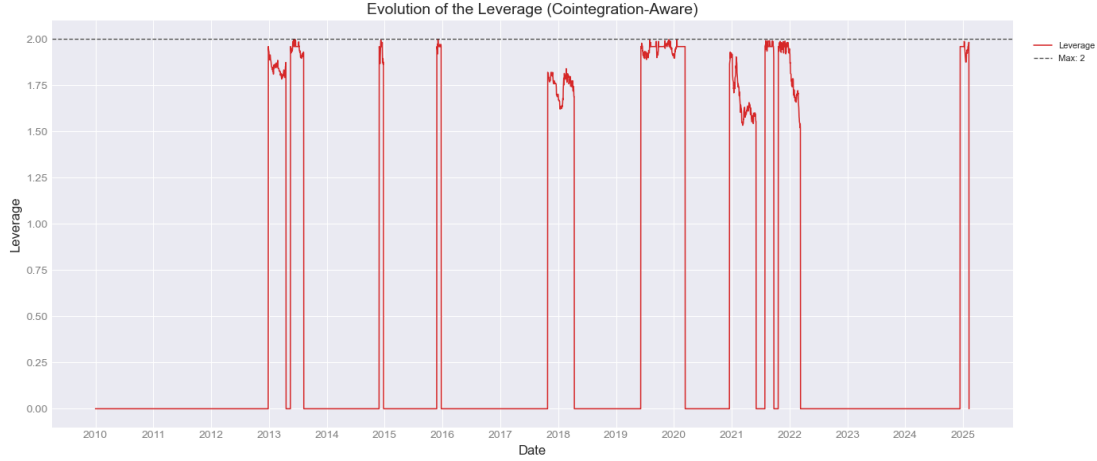


Figure 33: Evolution of the strategy's leverage over time under the adjusted cointegration break strategy. The leverage remains capped at 2, indicating adherence to the margin constraints even during periods when the cointegration relationship is absent.

$\alpha = 0.05$	
Metric	Value
Trading Session Start	2011-12-02
First Trade Date	2011-12-02
Last Trade Date	2025-02-07
Trading Session End	2025-02-07
p_threshold	0.05
z_in	NaN
z_stop	Not used
W_0	1000.00
W_max	1258.45
W_min	184.03
W_final	569.09
Maximum Leverage	2
Window Size (days)	500
Step Size (days)	20
Number of Windows	173
Net Profit	-430.91
Return (%)	-43.09
Nb trades	11
Max Drawdown (%)	-85.38
Max Leverage Used	2.00
First Bankruptcy Date	No Bankruptcy
Coint. Days	581
Coint. % of trading days	14.74
Coint. regime Changes	22
Exits due to Coint. Break	10
Test Type	ADF

Figure 34: Key Performance of the out-of-sample pairs pair trading strategy with entry and exit decisions are based on a cointegration condition (p-value < 0.05).

The detailed trade logs for Table 34 can be found in Table 26 in the Appendix⁴.

$\alpha = 0.01$	
Metric	Value
Trading Session Start	2011-12-02
First Trade Date	2017-10-27
Last Trade Date	2022-03-11
Trading Session End	2025-02-07
p_threshold	0.01
z_in	NaN
z_stop	Not used
W_0	1000.00
W_max	1382.44
W_min	936.89
W_final	1371.89
Maximum Leverage	2
Window Size (days)	500
Step Size (days)	20
Number of Windows	173
Net Profit	371.89
Return (%)	37.19
Nb trades	3
Max Drawdown (%)	-29.00
Max Leverage Used	1.99
First Bankruptcy Date	No Bankruptcy
Coint. Days	140
Coint. % of trading days	3.55
Coint. regime Changes	7
Exits due to Coint. Break	3
Test Type	ADF

Figure 35: Key Performance of the out-of-sample pairs pair trading strategy with entry and exit decisions are based on a cointegration condition (p-value < 0.01).

The detailed trade logs for Table 35 can be found in Table 27 in the Appendix⁴.

Tables 34 ($\alpha = 0.05$) and 35 ($\alpha = 0.01$), along with Figures 31, 32, and 33, illustrate the different outcomes of the adjusted strategy. Under the more lenient threshold ($\alpha = 0.05$), the strategy initiates 11 trades but ends with a net loss of \$430.91 (return of -43.09%) and a substantial drawdown of -85.38%. In contrast, the stricter threshold ($\alpha = 0.01$) triggers just 3 trades yet achieves a net profit of \$371.89 (return of 37.19%), with a markedly lower drawdown of -29.00%.

These results underscore the trade-off between generating a higher volume of trades based on less robust cointegration signals versus focusing on more selective but potentially more reliable relationships. While allowing for more trades can capture additional market opportunities, it also risks deeper drawdowns and weaker returns. Conversely, tightening the cointegration criteria reduces the frequency of trades but can improve risk-adjusted performance. Future refinements could involve adaptive thresholds or supplementary tests (e.g., incorporating error-correction models) to strike a more effective balance between capturing profitable signals and mitigating downside risks.

Q4.16

This study has shown that pair trading, grounded in cointegration and mean reversion, can identify potential opportunities in equity markets. We tested various thresholds, leverage levels, and stop-loss parameters with adjusted daily prices, revealing that while some strategies can generate positive returns in simulations, several critical factors limit their real-world viability.

First, we found that different strategies produce mixed results. Our tests demonstrated that modest leverage can yield respectable returns, but pushing leverage too far (for example, up to 20) significantly increases drawdowns and can occasionally lead to losses that exceed the initial capital. In addition, the slow mean-reversion nature of certain spreads complicates holding large or leveraged positions over extended periods.

Second, relying solely on adjusted daily closing prices overlooks real market microstructure, including intraday volatility, trading halts, and partial fills. In practice, substantial price divergences may occur within the same trading day, and ignoring them can generate overly optimistic results.

Third, transaction costs and liquidity constraints—such as commissions, bid-ask spreads, short-selling fees, and slippage—erode theoretical profits. These expenses become more severe with higher-frequency execution or larger positions, making profitability estimates potentially unrealistic unless such costs are carefully modeled and monitored.

Fourth, there is a time-frame concern and a “repainting” issue. Because we based signals on end-of-day data and then assumed entry or exit at a subsequent open, the signal that appeared valid at the prior close might not hold intraday. This risk intensifies if the price swiftly mean-reverts before the next session, invalidating the signal or altering the position’s risk profile.

Fifth, margin, stop-loss, and take-profit mechanics require careful attention. While margin magnifies gains, it also exacerbates drawdowns and demands robust margin-call management. Although stop-loss and take-profit orders can protect against extreme volatility, they may increase whipsaw risk if the spread temporarily widens before reverting. An overly tight stop-loss might close positions prematurely, whereas the absence of one can expose a trade to unlimited losses if a major dislocation persists.

Finally, focusing solely on cointegration and mean reversion neglects other dimensions of market dynamics, such as shifting volatility regimes, macroeconomic events, and company-specific news. Incorporating additional data and models—for instance, volatility forecasting, event studies, or fundamental ratios—is crucial for more robust decision-making.

Ultimately, while pair trading can offer market-neutral returns in theory, the interplay of realistic transaction costs, uncertain holding durations, high leverage, and operational challenges makes these strategies difficult to maintain as a primary hedge-fund driver. Success demands rigorous risk management, advanced statistical measures, robust operational protocols, and complementary trading approaches, ensuring that pair trading evolves from a compelling academic concept into a consistently reliable instrument for long-term portfolio performance.

Appendix

Complementary Audit Analysis and Tests

In light of the strong persistence observed in the normalized spread $\hat{z}_t$ (see Figure 9 and Table 10) and the significant results of the Ljung-Box test, we conducted additional tests to further investigate the stationarity of the spread between CHEVRON and CONOCOPHILLIPS.

We exploited the same method of the Monte Carlo simulations presented in Table 7 (with $T \approx 3941$, $N = 10\,000$, Seed = 123) to obtain simulated critical values and test the null hypothesis of non-stationarity on the spread (result in Table 19). We then confirmed these results by applying the `statsmodels.adfuller` function with `maxlag = 0` to ensure consistency with our simulation framework and to avoid the inclusion of additional lags inherent to the augmented version of the test (result in Table 20).

Furthermore, we used the `statsmodels.adfuller` function to verify the stationarity of TOTALENERGIES, thereby confirming its exclusion from the cointegration analysis (see Q2.8 and the Audit Table 21). Finally, an additional cointegration test was performed using the `coin` with `maxlag = 0` function from `statsmodels` (see the Audit Table 22) on all pairs to confirm the selection of the winning pair CHEVRON and CONOCOPHILLIPS.

1. Stationarity Test using Monte Carlo Simulation

DF_Statistic	Critical_10%	Conclusion_10%	Critical_5%	Conclusion_5%	Critical_1%	Conclusion_1%	p_value	T
-3.8568	-2.5682	Reject H_0 (Stationary)	-2.8759	Reject H_0 (Stationary)	-3.4461	Reject H_0 (Stationary)	0.0026	3941

Table 19: DF test results for the raw price spread between CHEVRON and CONOCOPHILLIPS using simulated critical values (simulation run with $T \approx 3941$, $N = 10\,000$, Seed = 123). The test statistic (DF = -3.8568) is lower than the critical values at the 10% and 5% significance levels (-2.5682 and -2.8759, respectively), leading to rejection of the null hypothesis of non-stationarity at these levels. However, at the 1% level (critical value = -3.4461), the null hypothesis is not rejected (p-value = 0.0106). Green cells indicate rejection of H_0 (suggesting stationarity).

2. Stationarity Test using `statsmodels.adfuller` function

ADF_Statistic	Critical_10%	Conclusion_10%	Critical_5%	Conclusion_5%	Critical_1%	Conclusion_1%	p_value
-3.8568	-2.5672	Reject H_0 (Stationary)	-2.8623	Reject H_0 (Stationary)	-3.4320	Reject H_0 (Stationary)	0.0024

Table 20: DF test results for the raw price spread between CHEVRON and CONOCOPHILLIPS obtained using the `statsmodels.adfuller` function with `maxlag = 0` (simulation run with $T \approx 3941$, $N = 10\,000$, Seed = 123). The ADF test statistic of -3.8568 is lower than the simulated critical values at the 10% (-2.5672) and 5% (-2.8623) levels, leading to rejection of the null hypothesis of non-stationarity (indicating stationarity) at these significance levels. At the 1% level (critical value = -3.4320), the null hypothesis is also rejected, with a p-value of 0.0024. Green cells indicate rejection of H_0 , thereby suggesting stationarity.

3. (Q2.8 Audit on the Results)

	ADF Statistic	p-value	Used Lags	Number of Observations	Critical Value (1%)	Critical Value (5%)	Critical Value (10%)	Conclusion
TOTALENERGIES	-4.6598	0.0001	0	3940	-3.4320	-2.8623	-2.5672	Stationary
CHEVRON	-2.5651	0.1004	0	3940	-3.4320	-2.8623	-2.5672	Non-stationary
CONOCOPHILLIPS	-1.9408	0.3131	0	3940	-3.4320	-2.8623	-2.5672	Non-stationary
EXXON_MOBIL	-1.8347	0.3634	0	3940	-3.4320	-2.8623	-2.5672	Non-stationary
OCCIDENTAL_PTL	-1.7945	0.3832	0	3940	-3.4320	-2.8623	-2.5672	Non-stationary

Table 21: This table confirms the results from Q2.8 (see Tables 4, 5, 6) indicating that TOTALENERGIES is stationary and should be excluded from our pair trading strategy. The DF test was performed with a constant term and with `maxlag = 0` to ensure consistency with the simulated critical value distribution.

4. (Q3.2 Audit on the Results)

We did an audit test in order to have confirmation for the right selection of winning pair for the pair trading strategy with `coin` from the python package `statsmodels.tsa.stattools`.

Asset1	Asset2	Direction	Test Statistic	p-value	Conclusion
CHEVRON	CONOCOPHILLIPS	CHEVRON → CONOCOPHILLIPS	-3.3888	0.0436	Cointegrated
CONOCOPHILLIPS	CHEVRON	CONOCOPHILLIPS → CHEVRON	-2.9664	0.1184	Not cointegrated
CHEVRON	EXXON_MOBIL	CHEVRON → EXXON_MOBIL	-2.3124	0.3674	Not cointegrated
CHEVRON	OCCIDENTAL_PTL	CHEVRON → OCCIDENTAL_PTL	-2.2781	0.3845	Not cointegrated
CONOCOPHILLIPS	EXXON_MOBIL	CONOCOPHILLIPS → EXXON_MOBIL	-1.7767	0.6410	Not cointegrated
OCCIDENTAL_PTL	CHEVRON	OCCIDENTAL_PTL → CHEVRON	-1.7444	0.6563	Not cointegrated
OCCIDENTAL_PTL	CONOCOPHILLIPS	OCCIDENTAL_PTL → CONOCOPHILLIPS	-1.6892	0.6818	Not cointegrated
CONOCOPHILLIPS	OCCIDENTAL_PTL	CONOCOPHILLIPS → OCCIDENTAL_PTL	-1.6664	0.6921	Not cointegrated
EXXON_MOBIL	CONOCOPHILLIPS	EXXON_MOBIL → CONOCOPHILLIPS	-1.6464	0.7009	Not cointegrated
EXXON_MOBIL	CHEVRON	EXXON_MOBIL → CHEVRON	-1.6061	0.7183	Not cointegrated
OCCIDENTAL_PTL	EXXON_MOBIL	OCCIDENTAL_PTL → EXXON_MOBIL	-1.3474	0.8150	Not cointegrated
EXXON_MOBIL	OCCIDENTAL_PTL	EXXON_MOBIL → OCCIDENTAL_PTL	-1.1974	0.8586	Not cointegrated

Table 22: The results confirm the conclusion from Q3.2 (see Table 8) that CHEVRON and CONOCOPHILLIPS constitute the winning pair for our pair trading strategy. The cointegration test was conducted using the `coint`⁵ function from `statsmodels.tsa.stattools`⁶, which applies a DF procedure with a constant term and `maxlag = 0`, ensuring consistency with our simulated critical values.

Half life and IRF

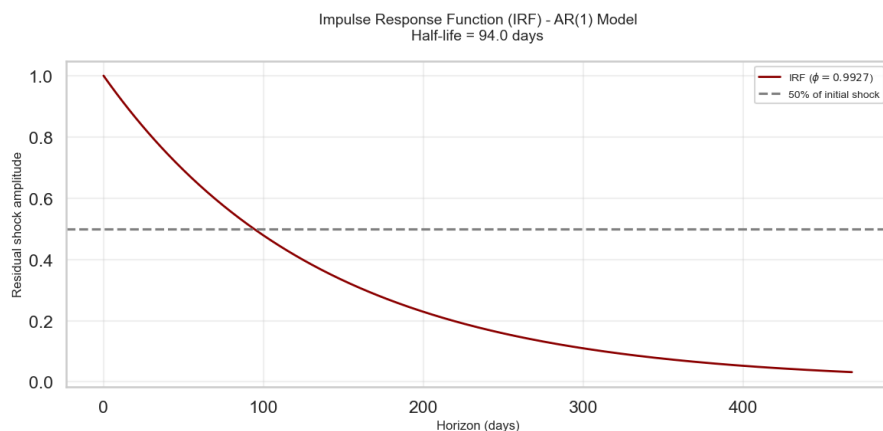


Figure 36: Impulse Response Function (IRF) of the spread. The IRF illustrates the dynamic response of the spread to a shock, highlighting its persistence over time.

AR(1) Model Summary:	Value
Mean of spread	0.0000
Variance of spread	89.5745
Phi coefficient (AR1)	0.9927
Half-life	94.0 days

Figure 37: Summary statistics of the AR(1) model fitted on the spread. The high ϕ coefficient and long half-life confirm the strong persistence observed in the ACF and Ljung-Box tests.

The impulse response function (Figure 36) demonstrates how the normalized spread reacts to a shock over time, while the AR(1) model summary (Table 37) quantifies this persistence with a ϕ coefficient of 0.9927 and a half-life of 94 days. These findings are consistent with the high autocorrelation values reported in Table 10 and the significant Ljung-Box test results, reinforcing the conclusion that the spread exhibits strong persistence. Such persistence has important implications for trading strategies, as it indicates that deviations from the mean may take a considerable amount of time to revert.

Q4.5 Detailed derivation of the Profit

We begin with the cointegration relationship:

$$P_t^A = \alpha + \beta P_t^B + z_t, \quad \text{so that } z_t = P_t^A - \alpha - \beta P_t^B.$$

The normalized spread is defined as

$$\tilde{z}_t = \frac{z_t}{\sigma(z_t)},$$

where $\sigma(z_t)$ is the standard deviation of z_t .

At the entry signal (time t_0), we have:

$$\tilde{z}_{t_0} = \tilde{z}^{\text{in}},$$

which implies

$$z_{t_0} = \tilde{z}^{\text{in}} \sigma(z_t).$$

At time t_0 , we open a position by short-selling 1 unit of asset A and buying β units of asset B . This position is given by

$$Q_1 = \begin{pmatrix} -1 \\ \beta \end{pmatrix}.$$

Thus, the net cash flow at entry is:

$$V_{\text{entry}} = -P_{t_0}^A + \beta P_{t_0}^B.$$

Using the cointegration relation at t_0 :

$$P_{t_0}^A = \alpha + \beta P_{t_0}^B + z_{t_0},$$

we substitute to obtain:

$$V_{\text{entry}} = -(\alpha + \beta P_{t_0}^B + z_{t_0}) + \beta P_{t_0}^B = -\alpha - z_{t_0}.$$

Assuming that the position is closed when the normalized spread reaches zero (i.e. $z_{t_{\text{exit}}} = 0$), we have by definition:

$$z_{t_{\text{exit}}} = P_{t_{\text{exit}}}^A - \alpha - \beta P_{t_{\text{exit}}}^B = 0.$$

This immediately implies:

$$P_{t_{\text{exit}}}^A = \alpha + \beta P_{t_{\text{exit}}}^B.$$

Substituting this result into the expression for the portfolio value at exit,

$$V_{\text{exit}} = -P_{t_{\text{exit}}}^A + \beta P_{t_{\text{exit}}}^B,$$

we obtain:

$$V_{\text{exit}} = -(\alpha + \beta P_{t_{\text{exit}}}^B) + \beta P_{t_{\text{exit}}}^B = -\alpha.$$

The profit from the trade is therefore:

$$\text{Profit} = V_{\text{exit}} - V_{\text{entry}} = (-\alpha) - (-\alpha - z_{t_0}) = z_{t_0}.$$

Since $z_{t_0} = \tilde{z}^{\text{in}} \sigma(z_t)$, it follows that:

$$\text{Profit} = \tilde{z}^{\text{in}} \sigma(z_t).$$

Q4.7 Detailed Trading logs - Strategy with $L = 2$

	Signal	Entry Date	Exit Date	Exit Reason	Profit	Return (%)	Entry Wealth	Exit Wealth
1	Signal 2	2010-01-25	2011-07-20	Signal crossover	591.00	59.00	1000.00	1591.00
2	Signal 2	2015-07-01	2016-02-09	Signal crossover	-783.00	-49.00	1591.00	808.00
3	Signal 1	2016-07-25	2018-08-17	Signal crossover	351.00	43.00	808.00	1159.00
4	Signal 1	2019-06-24	2020-01-22	Signal crossover	-119.00	-10.00	1159.00	1040.00
5	Signal 2	2020-03-19	2020-04-20	Signal crossover	590.00	57.00	1040.00	1630.00
6	Signal 2	2021-09-22	2022-02-28	Signal crossover	1089.00	67.00	1630.00	2719.00
7	Signal 1	2022-03-09	2022-09-02	Signal crossover	91.00	3.00	2719.00	2810.00
8	Signal 1	2022-12-30	2023-07-14	Signal crossover	-652.00	-23.00	2810.00	2158.00
9	Signal 2	2023-10-30	2024-07-31	Signal crossover	114.00	5.00	2158.00	2272.00
10	Signal 2	2024-08-30	2024-11-15	Signal crossover	201.00	9.00	2272.00	2473.00

Table 23: Detailed Logs of the Table 12 and Table 15 for the pair-trading strategy using $z_{in} = 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, and leverage constraint of 50% are reported at the end of the trading horizon.

Q4.8 Detailed Trading logs - Strategy with $L = 20$

	Signal	Entry Date	Exit Date	Exit Reason	Profit	Return (%)	Entry Wealth	Exit Wealth
1	Signal 2	2010-01-25	2011-07-20	Signal crossover	1285.04	128.50	1000.00	2285.04
2	Signal 2	2015-07-01	2016-02-09	Signal crossover	-2137.08	-93.52	2285.04	147.96
3	Signal 1	2016-07-25	2018-08-17	Signal crossover	642.19	434.01	147.96	790.15
4	Signal 1	2019-06-24	2020-01-22	Signal crossover	-813.10	-102.90	790.15	-22.94
5	Signal 2	2020-03-19	2020-04-20	Signal crossover	-23.48	-102.32	-22.94	-46.42
6	Signal 2	2021-09-22	2022-02-28	Signal crossover	-88.42	-190.48	-46.42	-134.84
7	Signal 1	2022-03-09	2022-09-02	Signal crossover	-81.00	-60.07	-134.84	-215.84
8	Signal 1	2022-12-30	2023-07-14	Signal crossover	858.06	397.54	-215.84	642.22
9	Signal 2	2023-10-30	2024-07-31	Signal crossover	129.38	20.15	642.22	771.60
10	Signal 2	2024-08-30	2024-11-15	Signal crossover	211.08	27.36	771.60	982.68

Table 24: Detailed Logs of the Table 13 for the pair-trading strategy using $z_{in} = 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, and leverage constraint of 500% are reported at the end of the trading horizon.

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Q4.13 Detailed Trading logs - Out of the Sample Strategy

	Signal	Entry Date	Exit Date	Exit Reason	Profit	Return (%)	Entry Wealth	Exit Wealth
1	Signal 1	2011-12-12	2012-03-21	Signal crossover	124.55	12.46	1000.00	1124.55
2	Signal 1	2012-04-30	2012-06-12	Signal crossover	-79.23	-7.05	1124.55	1045.32
3	Signal 1	2012-07-20	2012-11-14	Signal crossover	-98.81	-9.45	1045.32	946.51
4	Signal 1	2013-01-31	2013-06-19	Signal crossover	96.79	10.23	946.51	1043.30
5	Signal 2	2013-08-02	2013-12-19	Signal crossover	0.54	0.05	1043.30	1043.83
6	Signal 2	2014-02-05	2014-04-29	Signal crossover	326.78	31.31	1043.83	1370.61
7	Signal 2	2014-10-01	2015-10-30	Signal crossover	-673.68	-49.15	1370.61	696.93
8	Signal 1	2015-12-15	2017-10-27	Signal crossover	206.79	29.67	696.93	903.71
9	Signal 2	2018-02-01	2019-03-01	Signal crossover	52.55	5.82	903.71	956.27
10	Signal 1	2019-07-24	2019-08-15	Signal crossover	-187.20	-19.58	956.27	769.06
11	Signal 2	2020-01-24	2020-11-10	Signal crossover	-480.36	-62.46	769.06	288.71
12	Signal 2	2021-02-16	2021-04-20	Signal crossover	32.97	11.42	288.71	321.68
13	Signal 2	2021-05-18	2021-12-17	Signal crossover	71.57	22.25	321.68	393.25
14	Signal 1	2022-03-01	2022-06-07	Signal crossover	175.26	44.57	393.25	568.51
15	Signal 1	2022-07-29	2022-08-18	Signal crossover	10.41	1.83	568.51	578.93
16	Signal 2	2022-09-02	2022-11-23	Signal crossover	181.62	31.37	578.93	760.55
17	Signal 1	2023-04-18	2023-05-18	Signal crossover	-112.89	-14.84	760.55	647.65
18	Signal 2	2023-07-20	2024-05-10	Signal crossover	92.62	14.30	647.65	740.28
19	Signal 2	2024-08-07	2024-11-13	Signal crossover	124.31	16.79	740.28	864.59

Table 25: Detailed Logs of the Table 18 using $z_{in} = 1.50$ and an initial wealth of \$1,000. Final wealth, drawdowns, margin constraint of 50 % for the out of sample pairs trading strategy using rolling window estimation with a window size of 500 days and step size of 20 days

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Q4.15 Detailed Trading logs - OOS Strategy - Cointegration test with $\alpha = 0.05$

	Signal	Entry Date	Exit Date	Exit Reason	Profit	Return (%)	Entry Wealth	Exit Wealth	P-value
1	Signal 1	2011-12-02	2011-12-30	Cointegration break	62.57	6.26	1000.00	1062.57	0.05
2	Signal 2	2012-12-28	2013-02-22	Cointegration break	136.78	12.87	1062.57	1199.35	0.04
3	Signal 1	2013-03-22	2013-04-19	Cointegration break	-112.06	-9.34	1199.35	1087.29	0.02
4	Signal 1	2013-05-17	2013-07-12	Cointegration break	24.42	2.25	1087.29	1111.71	0.04
5	Signal 2	2014-10-31	2014-11-28	Cointegration break	-194.80	-17.52	1111.71	916.91	0.01
6	Signal 2	2015-11-27	2015-12-25	Cointegration break	-38.32	-4.18	916.91	878.58	0.02
7	Signal 2	2017-10-27	2018-02-16	Cointegration break	18.76	2.14	878.58	897.34	0.00
8	Signal 2	2018-03-16	2018-04-13	Cointegration break	141.45	15.76	897.34	1038.79	0.03
9	Signal 1	2019-06-07	2020-03-13	Cointegration break	-738.00	-71.04	1038.79	300.79	0.03
10	Signal 2	2021-11-19	2022-03-11	Cointegration break	285.74	95.00	300.79	586.53	0.02
11	Signal 1	2024-12-13	2025-02-07	End of sample	-17.44	-2.97	586.53	569.09	0.04

Table 26: Detailed Logs of the Table Table 34 presents the out-of-sample pairs trading strategy. It starts with an initial wealth of 1000 and applies a 50% margin constraint. The strategy uses a rolling window estimation with a 500-day window and a 20-day step, while entry and exit decisions are based on a cointegration condition (p-value < 0.05, the numbers on the table are rounded to 2 decimals).

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Q4.15 Detailed Trading logs - OOS Strategy - Cointegration test with $\alpha = 0.01$

	Signal	Entry Date	Exit Date	Exit Reason	Profit	Return (%)	Entry Wealth	Exit Wealth	P-value
1	Signal 2	2017-10-27	2018-02-16	Cointegration break	21.35	2.14	1000.00	1021.35	0.00
2	Signal 2	2019-11-22	2020-01-17	Cointegration break	21.73	2.13	1021.35	1043.08	0.01
3	Signal 1	2022-02-11	2022-03-11	Cointegration break	328.81	31.52	1043.08	1371.89	0.01

Table 27: Detailed Logs of the Table Table 34 presents the out-of-sample pairs trading strategy. It starts with an initial wealth of 1000 and applies a 50% margin constraint. The strategy uses a rolling window estimation with a 500-day window and a 20-day step, while entry and exit decisions are based on a cointegration condition (p-value < 0.01, the numbers on the table are rounded to 2 decimals).

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